Molecular Dynamics Simulations of Bacterial Actin Protein MreB



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Abstract

This dissertation investigates the conformational and mechanical behaviour of the bacterial actin homologue MreB from the Gram-positive bacterium *Geobacillus stearothermophilus* using all-atom molecular dynamics simulations. The simulated structures are analysed using a reduced description based on the relative motion of four structural subdomains. We then compare the resulting conformational behaviour across different nucleotide, polymerisation and membrane conditions. Particular attention is given to the effect of membrane association on conformational flexibility and on the stability of the interface between neighbouring MreB subunits. We also develop a new membrane-referenced coarse-grained mechanical model of $MreB^{Gs}$ using information extracted from the atomistic trajectories. The model incorporates both protein interactions and membrane coupling, and is validated against an independent collective motion.

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Chapter 1

Introduction

Cytoskeletal proteins assemble into filaments that perform important structural and mechanical roles within cells. One of the best studied examples is actin, a protein found in eukaryotic (animal and plant) cells whose filaments contribute to cell shape, movement and many other essential cellular processes [1]. Bacteria possess a structurally related protein called MreB, often described as a bacterial homologue of actin. Although MreB is known to form filaments and to play an important role in bacterial cell shape [2], considerably less is understood about how these filaments are formed and how their behaviour is controlled [3].

Many fundamental questions about MreB therefore remain open. These include how its filaments are built and how their arrangement within the cell is controlled. It is also unclear whether MreB behaves differently between bacterial species and, if so, what causes these differences. The recent determination of the first three-dimensional structures of MreB from the Gram-positive bacterium *Geobacillus stearothermophilus* ($MreB^{Gs}$) provides a new opportunity to investigate these questions [3]. The main approach used in this dissertation is all-atom molecular dynamics (MD) simulation, which allows the motion and changing structure of $MreB^{Gs}$ to be studied at atomic resolution.

Understanding MreB is important because it is closely connected to the processes responsible for constructing the bacterial cell wall [4]. This structure is essential for bacterial survival and is already the target of several major classes of antibiotics [5]. Disruption of MreB can interfere with normal cell-wall organisation and produce severe changes in cell shape [6]. Developing a clearer molecular understanding of MreB may therefore contribute both to our understanding of bacterial growth and to the longer-term search for new ways of disrupting it [7]. More broadly, studying

MreB can help reveal how the behaviour of individual proteins contributes to the organisation of bacterial cells.

1.1 Main Contributions

The main contributions of this dissertation are summarised below.

1. **The first MD investigation of a Gram-positive MreB homologue.**

   Previous all-atom MD studies of MreB have focused exclusively on homologues from Gram-negative bacteria. This work provides the first MD investigation of a Gram-positive MreB homologue, using the recently determined structures of $MreB^{Gs}$. The results reveal both similarities to previously studied Gram-negative homologues and important differences in the conformational and membrane-associated behaviour of $MreB^{Gs}$.

2. **A reproducibility-focused analysis of MreB dynamics.**

   This work argues that replica-based analysis is essential for drawing reliable conclusions from MreB simulations. Previous studies typically drew conformational conclusions from only one or two independent trajectories, whereas five replicas are used here for each condition. This greater emphasis on reproducibility provides a more rigorous basis for distinguishing consistent molecular behaviour from variation between individual trajectories.

3. **A new membrane-coupled coarse-grained mechanical model of MreB.**

   This work develops a bespoke coarse-grained mechanical model for MreB. Although inspired by earlier coarse-grained models of actin, it is redesigned to include features that are central to MreB, particularly its interaction with the membrane. The model is informed by structural insights from the all-atom simulations, and its parameters are obtained directly from the atomistic trajectories. Here, the model is parameterised using $MreB^{Gs}$ data, but the same framework can be applied directly to other MreB homologues using the corresponding simulation data.

1.2 Structure of the Dissertation

The remainder of this dissertation is organised as follows. Chapter 2 provides the biological and computational background, introduces the $MreB^{Gs}$ systems studied, and

describes their preparation and MD simulations. Chapter 3 analyses the resulting trajectories, focusing on the conformational behaviour of $MreB^{Gs}$, reproducibility across independent simulations, and the effects of membrane association. Chapter 4 uses these atomistic results to develop, parameterise and validate a new membrane-coupled coarse-grained model of MreB. Finally, Chapter 5 discusses the main findings, their limitations and directions for future work. A table defining key biological terminology used throughout the dissertation is provided below.

Term	**Definition**
Nucleotide	A small molecule that can bind to a protein and influence its behaviour. In this work, the relevant nucleotides are ATP and ADP.
Polymerisation	The process by which individual protein molecules join together to form a larger structure. The individual protein molecules within such an assembly are referred to as subunits.
Monomer	A single protein molecule that is not joined to another protein molecule.
Dimer	A structure formed by the association of two protein subunits.
Protofilament	A linear chain of protein subunits joined through polymerisation. Two MreB protofilaments can associate to form a double filament.
PDB ID	A unique identifier assigned to an experimentally determined molecular structure deposited in the Protein Data Bank.
Residue	A single amino acid within a protein chain. Each residue is made up of several atoms, and residues are numbered according to their position in the protein sequence.

Table 1.1: Definitions of key biological terminology used throughout this dissertation.

Chapter 2

Background and Preparation

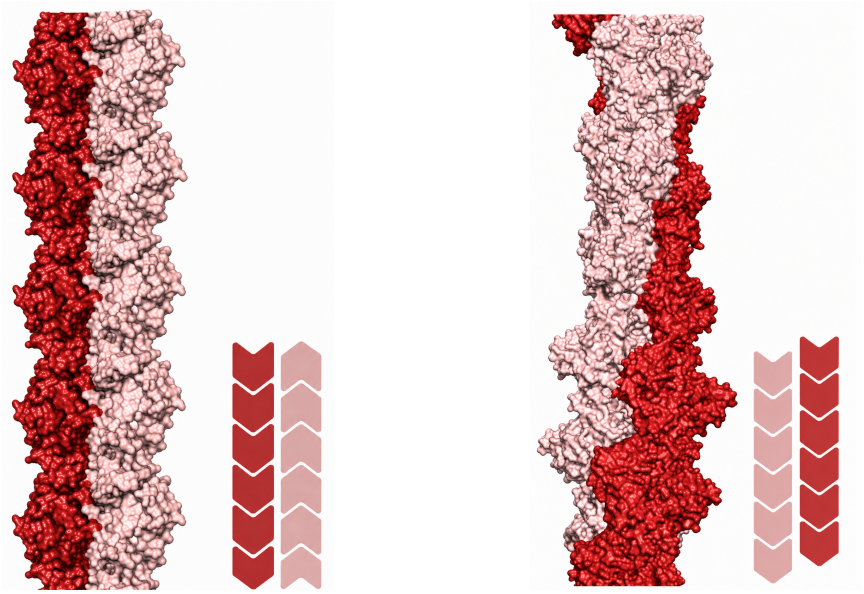
This chapter provides the background and preparation required before presenting the results in later chapters. It first reviews the biological understanding of MreB and previous MD studies of related MreB structures. The $MreB^{Gs}$ systems used in this work are then introduced and prepared for simulation. Finally, the MD methodology and simulation protocol used to generate the trajectories analysed in subsequent chapters are described.

2.1 Biological Background

Actin filaments form part of the cytoskeleton, a network of protein filaments that helps organise the cell, maintain its shape, and support many essential cellular processes [1]. Cytoskeletal proteins such as actin were traditionally thought to be absent from prokaryotes, including bacteria, and the cytoskeleton was therefore considered a uniquely eukaryotic feature. This view changed around 2001, when structural and cellular studies established MreB as a bacterial actin homologue and demonstrated that bacteria possess organised cytoskeletal structures [2]. MreB forms filaments that help organise cell-wall synthesis and plays a central role in maintaining the rod shape of many bacterial species [8].

Although MreB is structurally related to actin, the two proteins form distinct filament architectures. Actin forms twisted, polar filaments in which the two strands are oriented in the same direction. MreB instead forms relatively straight double filaments composed of two protofilaments oriented in opposite directions [9]. These contrasting filament architectures are illustrated schematically in Figure 2.1. A further distinction is that MreB can bind directly to membranes, whereas actin does not

normally associate with membranes without the help of other proteins [10].



(a) MreB filament **(b)** Actin filament

Figure 2.1: Schematic representations of MreB and actin filament architectures adapted from Wagstaff et al. [11]. (a) MreB forms relatively straight antiparallel double filaments, with the two protofilaments oriented in opposite directions. (b) Actin forms a twisted polar filament in which the two strands are oriented in the same direction. The arrows indicate the relative orientation of the two strands in each filament.

A further important feature of both actin and MreB is their interaction with nucleotides, particularly ATP and ADP. The identity of the bound nucleotide can influence both protein structure and polymerisation. ATP can be converted to ADP through hydrolysis, linking the two nucleotide states and motivating their comparison throughout this work. In actin, the influence of nucleotide state is well established. Both ATP- and ADP-bound actin can polymerise, although nucleotide state affects polymerisation and filament stability [12]. ATP hydrolysis usually occurs after polymerisation and promotes later filament disassembly [13]. In MreB, however, the coupling between nucleotide state, hydrolysis and polymerisation is less clear and appears to vary between homologues [2, 14, 15].

Bacteria are commonly grouped into Gram-positive and Gram-negative categories based on differences in the structure surrounding the cell. Early structural and biochemical studies of MreB focused mainly on homologues from Gram-negative bacteria, including *Thermotoga maritima* (MreB^{Tm}), *Escherichia coli* (MreB^{Ec}) and *Caulobacter crescentus* (MreB^{Cc}) [2, 9, 10]. These proteins therefore formed the basis of much

of the current structural understanding of MreB. Gram-positive MreB proved more difficult to study experimentally because of its tendency to aggregate during purification. In 2023, Mao et al. reported the first three-dimensional structure of MreB from a Gram-positive bacterium, *Geobacillus stearothermophilus* ($MreB^{Gs}$) [3].

The polymerisation behaviour of $MreB^{Gs}$ differed from that previously reported for several Gram-negative homologues. *In vitro* studies showed that some Gram-negative MreB proteins could form double filaments both in solution and on lipid surfaces [9,10]. In contrast, Mao et al. observed no clear double-filament formation for $MreB^{Gs}$ in solution. Filament formation instead occurred on lipid surfaces and was strongly promoted by ATP, whereas ADP produced little polymerisation [3]. These observations suggest that nucleotide state, polymerisation and membrane association are closely linked in $MreB^{Gs}$. Understanding how these three conditions influence the structure and dynamics of $MreB^{Gs}$ provides the central biological motivation for the simulations in this dissertation.

2.2 Previous Molecular Dynamics Studies

All-atom MD simulations have long been used to investigate eukaryotic actin, providing insight into conformational changes associated with nucleotide state and polymerisation [16–18]. Similar methods have subsequently been applied to MreB to investigate the effects of nucleotide state, polymerisation and membrane association [19–23]. These studies have focused exclusively on MreB homologues from Gram-negative bacteria. An important aim of this work is therefore to determine whether their conclusions are transferable to the recently characterised Gram-positive homologue $MreB^{Gs}$.

The first all-atom MD study of MreB examined $MreB^{Tm}$ [19]. ATP-bound monomers were generally found to adopt a more open and less flattened conformation than ADP-bound monomers, while polymerisation promoted flattening and closure of the MreB subunits. These broad nucleotide-dependent trends were reproduced in later simulations [20], although the precise conformations differed between the two studies.

Shi et al. later extended these simulations to $MreB^{Cc}$ [21]. ATP-bound monomers were again generally more open than ADP-bound monomers, and polymerisation promoted closure of the nucleotide-binding pocket. However, the degree of flattening showed the opposite nucleotide-dependent trend to that previously reported for $MreB^{Tm}$. This

suggested that some conformational responses to nucleotide state may be homologue-specific.

The same study also examined MreB double filaments in solution and in association with a membrane. In solution, ATP-bound filaments showed greater twisting than ADP-bound filaments, while membrane binding reduced this twisting and stabilised the double-filament structure [21]. Later microsecond-scale simulations showed that MreB filaments can move between several long-lived conformational states, with some states accessible under both nucleotide conditions [22]. These results suggested that nucleotide state may alter the relative populations of accessible conformations rather than determine a single unique structure. These later results therefore highlighted the importance of simulation length and independent replicas when interpreting MreB dynamics, an issue considered further in Section 3.2.1.

Previous MD studies of membrane-associated MreB began from preassembled filaments and therefore examined how the membrane affects an existing polymer, rather than whether membrane association promotes polymerisation [21]. This distinction is particularly important for MreB^{Gs}, where experimental work indicates that filament formation occurs preferentially on lipid surfaces [3]. The role of membrane association in stabilising the interactions required for MreB^{Gs} filament formation therefore remains an important question explored further in this work.

2.3 MreB^{Gs} Structures and Simulation Systems

This section describes the construction of the MreB^{Gs} systems used for simulation in this work. As discussed above, the simulations were designed to examine three factors: nucleotide state, polymerisation and membrane association. The experimental starting structure is first described, together with the structural regions that are important for the later analysis. The reconstruction of unresolved residues and the construction of the nucleotide-bound, dimeric and membrane-associated systems are then outlined.

2.3.1 Experimental MreB^{Gs} structures

The MreB^{Gs} systems used for simulation were constructed from the nucleotide-free protofilament structure determined by Mao et al. [3] (PDB ID 7ZPT). Other experimentally determined MreB^{Gs} structures include the ADP- and ATP-bound monomers

(PDB IDs 8AAM and 8AZG, respectively). These structures were not used as starting structures for the simulations, but are used later for comparison.

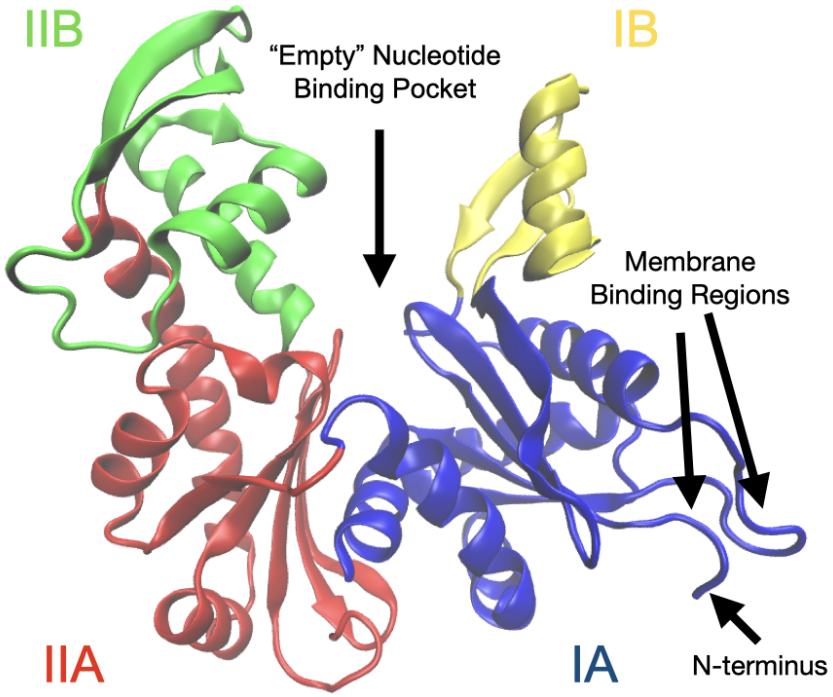


Figure 2.2: Crystal structure of nucleotide-free MreB^{Gs} (PDB ID 7ZPT) determined by Mao et al. [3], coloured by its four structural subdomains. Subdomain IA (blue) comprises residues 1–33, 68–145 and 314–340; IB (yellow) residues 34–67; IIA (red) residues 146–181 and 246–313; and IIB (green) residues 182–245. The nucleotide-binding pocket lies between the subdomains and is empty in this structure. The N-terminal region and hydrophobic $\alpha 2$–$\beta 7$ loop, both involved in membrane binding, are also indicated.

Figure 2.2 shows the experimental nucleotide-free structure of MreB^{Gs} (PDB ID 7ZPT). MreB adopts a four-subdomain architecture, conventionally divided into IA, IB, IIA and IIB. This four-subdomain description is widely used in the actin and MreB literature to characterise their relative arrangement and conformational motion [2, 24, 25]. The subdomain boundaries used here follow those defined for MreB^{Gs} by Mao et al. [3]. The nucleotide-binding pocket lies between these subdomains and is empty in the nucleotide-free 7ZPT structure.

Figure 2.2 also highlights two regions involved in membrane binding: the N-terminal region (residues 1–9) and the $\alpha 2$–$\beta 7$ loop (residues 91–101). Both regions are hydrophobic, tending to avoid water and favouring association with the interior of the

lipid membrane. The $\alpha 2$–$\beta 7$ loop is therefore referred to as the *hydrophobic loop* throughout this dissertation. Mao et al. showed that deletion of either region reduced membrane binding, with the strongest effect observed when both were removed [3].

2.3.2 Resolving missing residues

The experimental structure (PDB ID 7ZPT) contains two unresolved regions: an internal loop (residues 70–73) and a portion of the N-terminus (residues 1–3) [3]. Missing residues are common in experimentally determined protein structures, particularly in flexible regions that are difficult to resolve. For MD simulations, however, internal gaps are important to reconstruct, since an incomplete protein chain can introduce unrealistic local interactions and affect the subsequent dynamics [26]. Reconstruction of residues 70–73 was therefore particularly important before simulation.

The missing residues were reconstructed using MODELLER, a protein-structure modelling program based on the satisfaction of spatial restraints [27, 28]. For unresolved loops, MODELLER samples possible conformations of the missing region and evaluates their structural plausibility [26]. Twenty candidate structures were generated and ranked using the Discrete Optimized Protein Energy (DOPE) score [29], with the lowest-scoring model selected for the subsequent simulations. Other loop-modelling approaches are available, but for a short four-residue gap MODELLER provides a suitable and well-established reconstruction method [30].

The unresolved N-terminal residues 1–3 were reconstructed using the same procedure. Unlike residues 70–73, these terminal residues were not required to restore continuity of the protein chain and were therefore less important for simulation stability. They were nevertheless included because the N-terminal region contributes to membrane binding in $MreB^{Gs}$ [3]. Since the N-terminus is particularly flexible and is constrained by the resolved structure on only one side, its precise starting coordinates are less well determined. The reconstruction should therefore be regarded as providing a plausible starting geometry rather than a uniquely defined experimental conformation.

2.3.3 Construction of simulation systems

As shown in Figure 2.2, the nucleotide-binding pocket of the structure (PDB ID 7ZPT) is empty. ATP- and ADP-bound systems therefore had to be constructed before simulation. Although ATP- and ADP-bound $MreB^{Gs}$ structures are available experimentally (PDB IDs 8AZG and 8AAM), neither contains the Mg^{2+} ion that

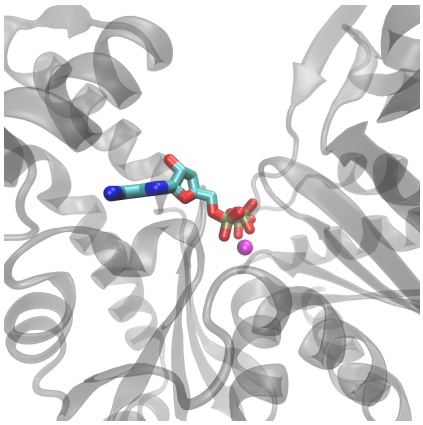
normally coordinates and stabilises the bound nucleotide. ATP together with its coordinating Mg^{2+} ion was therefore transferred from the related MreB structure 4CZM [9]. The structures were first aligned in VMD, after which the ATP and Mg^{2+} coordinates were placed into the 7ZPT binding pocket, as shown in Figure 2.3a. The corresponding ADP-bound system was then generated by removing the ATP γ-phosphate.

To investigate polymerisation, dimers were constructed from two neighbouring monomers along the experimentally resolved 7ZPT protofilament. The resulting system contains a single longitudinal intermonomer interface, with the two monomers referred to as the $(-)$ and $(+)$ subunits. Both ATP- and ADP-bound dimers were constructed. The lateral contacts between the two protofilaments forming the experimentally observed double filament (Figure 2.1a) were not structurally resolved and are therefore not represented in these simulations [3].

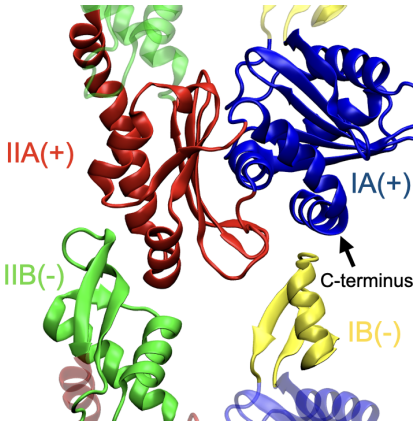
The longitudinal interface is illustrated in Figure 2.3b, where the principal contacting subdomains are labelled. It contains two main contacts: IIB of the $(-)$ subunit with IIA of the $(+)$ subunit, and IB of the $(-)$ subunit with IA of the $(+)$ subunit [3]. The latter contact is a distinctive feature of $MreB^{Gs}$. Here, subdomain IB of the $(-)$ subunit lies sufficiently far outward to contact subdomain IA of the neighbouring $(+)$ subunit, particularly its C-terminal region.

This positioning gives polymerised $MreB^{Gs}$ a more open conformation than that observed in several other MreB homologues. In more closed structures, such as $MreB^{Tm}$, subdomain IB instead lies closer to subdomain IIA of the neighbouring subunit [3]. The C-terminal region considered here comprises residues 330–340 and is labelled in Figure 2.3b.

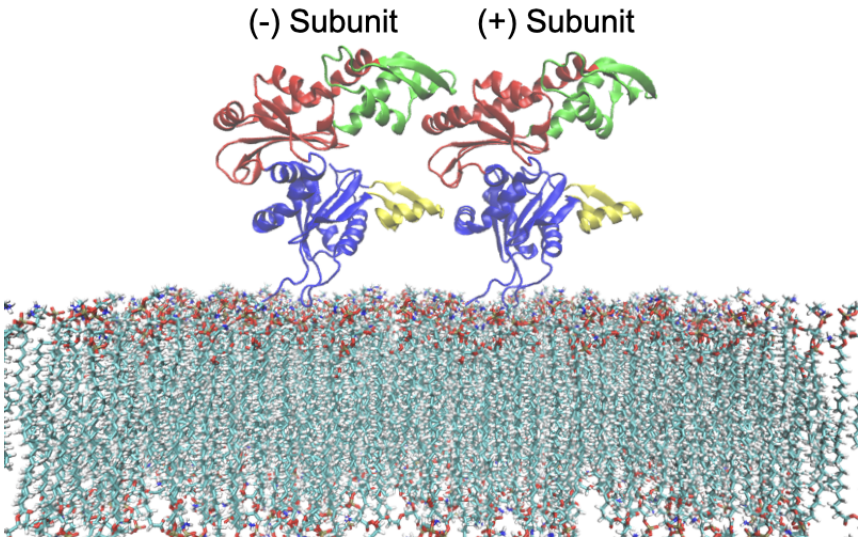
The final system consisted of an ATP-bound dimer associated with a lipid membrane. A phosphatidylethanolamine membrane patch composed of POPE lipids was generated using the Membrane plugin in VMD, following previous MreB simulations [21]. The dimer was oriented with the N-terminal region and hydrophobic loop facing the membrane, consistent with their experimentally identified roles in membrane binding [3]. These regions were initially positioned approximately 5 Å above the upper membrane phosphate plane, as shown in Figure 2.3c.



(a) ATP–Mg^{2+} binding pocket.



(b) Longitudinal interface



(c) Membrane-bound ATP dimer

Figure 2.3: Construction of the simulated $MreB^{Gs}$ systems. (a) ATP and its coordinating Mg^{2+} ion positioned within the nucleotide-binding pocket. ATP is shown in multicoloured stick representation, the protein in grey and Mg^{2+} as a pink sphere. (b) Longitudinal interface between the (−) and (+) subunits, showing the principal subdomain contacts and the C-terminal region of the (+) subunit. (c) Initial configuration of the membrane-bound ATP dimer, with the (−) and (+) subunits positioned above the membrane.

2.4 Molecular Dynamics Methodology

Experiments can provide detailed structural snapshots of biomolecules; however, these structures are inherently static. MD simulations complement such experiments by allowing their motion, deformation and interactions to be studied over time. The purpose of MD is not simply to generate molecular animations, but to obtain quantitative

information about properties such as molecular structure, flexibility, intermolecular interactions and the relative populations of different conformational states. This section first introduces the mathematical and computational principles underlying an MD trajectory, before describing how these methods were applied to the MreB^{Gs} systems constructed in Section 2.3.

2.4.1 Molecular dynamics theory

In classical all-atom MD, a system of N atoms is described by the vectors $\mathbf{q}(t), \mathbf{v}(t) \in \mathbb{R}^{3N}$, where $\mathbf{q}(t)$ and $\mathbf{v}(t)$ denote the atomic positions and velocities, respectively. Electron motion is not modelled explicitly. Instead, using the Born–Oppenheimer approximation, the interactions between atoms are represented by an empirical potential-energy function $U(\mathbf{q})$. The force acting on atom i is then obtained from the gradient of this potential,

$$\mathbf{F}_i(\mathbf{q}) = -\nabla_{\mathbf{q}_i} U(\mathbf{q}),$$

The form of $U(\mathbf{q})$ determines how the atoms interact. In an empirical force field, the potential energy is divided into bonded and non-bonded contributions. Bonded terms describe bond stretching, bond-angle bending and rotation about dihedral angles, while non-bonded terms describe van der Waals and electrostatic interactions. The total potential energy may therefore be written schematically as

$$U(\mathbf{q}) = \sum_{i=1}^{N_{\mathrm{bond}}} U_{\mathrm{bond},i} + \sum_{i=1}^{N_{\mathrm{angle}}} U_{\mathrm{angle},i} + \sum_{i=1}^{N_{\mathrm{dihedral}}} U_{\mathrm{dihedral},i} + \sum_{i=1}^{N_{\mathrm{nb}}} U_{\mathrm{nb},i}. \tag{2.1}$$

The functional forms and parameters associated with these terms define the force field and provide an approximate but computationally practical description of the molecular interactions.

Once the forces are known, the motion of each atom is governed by Newton's equations,

$$\frac{d\mathbf{q}_i}{dt} = \mathbf{v}_i, \tag{2.2}$$

$$m_i \frac{d\mathbf{v}_i}{dt} = -\nabla_{\mathbf{q}_i} U(\mathbf{q}), \tag{2.3}$$

where m_i is the mass of atom i. These equations cannot generally be solved analytically for a large molecular system and are therefore integrated numerically over a sequence of short time steps to generate a trajectory. Because atomic motions occur on very short time scales, MD simulations typically use an integration timestep

Δt on the order of femtoseconds (10^{-15} s). Verlet integration is commonly used to advance the atomic positions and velocities over successive time steps [31]. The prepared molecular structure provides the initial atomic configuration $\mathbf{q}(0)$, while the corresponding velocities $\mathbf{v}(0)$ are typically generated randomly from the Maxwell–Boltzmann distribution at the chosen temperature.

All-atom MD simulations can only represent a finite region of the surrounding solution. Without a suitable boundary treatment, atoms near the edge of this region would interact with an artificial vacuum. Periodic boundary conditions are commonly used to avoid this problem by treating the simulation cell as being surrounded by identical copies of itself. When an atom leaves through one side of the cell, its periodic image enters through the opposite side. This allows the finite simulation box to approximate part of a larger bulk environment.

The use of periodic boundaries also requires suitable treatment of non-bonded interactions, since atoms can interact with particles in neighbouring periodic images. Short-range interactions are commonly evaluated within a finite cut-off distance, while long-range electrostatics are treated using the particle-mesh Ewald method [32].

In practice, the equations of motion in Equations (2.2) and (2.3) are often modified to maintain the system close to chosen thermodynamic conditions. Temperature can be controlled using Langevin dynamics, which adds damping and random thermal forces to the deterministic atomic forces. Pressure can similarly be controlled by allowing the dimensions of the periodic simulation cell to vary around a chosen target value.

The MD calculations in this work were performed using NAMD [33], a software package for carrying out all-atom MD simulations. NAMD implements the force evaluation, numerical integration, periodic boundary conditions and thermodynamic control described above. The specific simulation conditions and protocol used for the MreB^{Gs} systems are described in the following section.

2.4.2 Simulation protocol

To carry out the MD simulations described in Section 2.4.1, a starting molecular structure is first required. We begin with the MreB^{Gs} systems constructed in Section 2.3. Each system was then prepared in VMD by surrounding the protein, or protein–membrane complex, with a box of water using the Solvate plugin. This represents the solvent environment surrounding the protein in a biological system. Approximately 15 Å of solvent padding was added around the system, with minor ad-

justments for the membrane-bound case. The resulting systems were then neutralised using the Autoionize plugin, which added Na^+ or Cl^- ions as required to remove the net charge. These steps produced the complete initial configurations used for the subsequent minimisation and MD simulations.

As described in Section 2.4.1, the force field assigns a potential energy $U(\mathbf{q})$ to each atomic configuration. Before MD, each prepared system was energy-minimised to remove unfavourable local interactions that may arise from the experimental structure or during system preparation. Such interactions can produce large atomic forces and numerical instability if dynamics are started immediately. Minimisation changes the atomic coordinates towards a nearby local minimum of $U(\mathbf{q})$. It does not represent physical time or thermal sampling and is therefore not included as part of the production trajectory. In this work, each system was minimised in NAMD for 10,000 steps using a conjugate-gradient method before simulation.

For the production simulations, the functional forms and parameters associated with the force-field terms in Equation (2.1) were specified by the CHARMM36 force field [34], while water molecules were represented using the TIP3P model [35]. For each system, NAMD was provided with a PSF file containing information about the atoms in the system, including their types, masses, charges, bonds, angles and dihedral angles. The equations of motion were integrated using $\Delta t = 2\,\mathrm{fs}$. Temperature was maintained at 310 K using Langevin dynamics with a damping coefficient of $1\,\mathrm{ps}^{-1}$, while pressure was maintained at approximately 1 atm using the Langevin piston method [36].

Five independent 140 ns replicas were performed for each simulated system, with each replica initiated independently from the same prepared structure. Previous MreB studies of monomers and dimers typically used only one or two replicas, with simulation lengths of approximately 55–140 ns and often below 100 ns [19–21]. The five replicas used here therefore provide a stronger basis for assessing the reproducibility of the observed molecular behaviour. Where discrete trajectory data are analysed, n_t denotes the number of saved trajectory frames included in the analysis. A summary of the simulated systems is provided in Appendix A.1.

Chapter 3

Molecular Dynamics Results

This chapter presents the main results obtained from analysis of the MD simulations. The high-dimensional trajectories are reduced to a small set of physically interpretable conformational variables, providing a basis for comparing the simulated systems. We then use this description to compare independent trajectories across nucleotide states, polymerisation states and membrane conditions, with particular emphasis on assessing which observed differences are reproducible rather than relying on individual simulations. This provides a more systematic basis for interpreting MreB conformational behaviour than in previous simulation studies. Finally, the effect of membrane association is examined in greater detail, providing new molecular insight into its role in the behaviour of polymerised $MreB^{Gs}$.

3.1 Reduced Description of $MreB^{Gs}$ Dynamics

All-atom MD trajectories contain the positions of thousands of atoms, making direct interpretation of the full conformational space impractical. It is therefore useful to describe the dominant protein motions using a much smaller number of collective coordinates. Previous studies of actin and MreB have done this by representing the protein through the centres of mass of its four structural subdomains and analysing their relative motion [16, 19, 20]. Before applying the same description to $MreB^{Gs}$, however, it is important to determine how well these subdomains actually represent the motions observed in the atomistic trajectories.

The following analysis therefore examines where the dominant motion occurs and how $MreB^{Gs}$ can most reasonably be reduced to a small number of collective variables. We ask whether the four visually defined subdomains shown in Figure 2.2 remain

approximately rigid, whether the important conformational changes occur primarily between rather than within them, and whether the conventional IA, IB, IIA and IIB partition captures the dominant collective dynamics. These questions are examined using the five independent ATP-bound monomer replicas as a representative subset of the full dataset, with the aim of identifying the general features of $\mathrm{MreB}^{\mathrm{Gs}}$ motion.

3.1.1 Flexible regions obscure subdomain dynamics

With the subdomains defined as in Figure 2.2, two broad types of motion can contribute to the observed conformational changes. The first is rigid-body motion, in which a subdomain moves as a whole relative to the other structural regions. The second is internal deformation within the subdomain itself. Distinguishing between these is important because the reduced analysis is mainly concerned with large-scale relative motion between the subdomains, which is captured by rigid-body motion. By contrast, internal deformation can be dominated by highly flexible local regions whose fluctuations may not reflect the overall conformational behaviour of the protein.

To separate these contributions, we first need a way of comparing protein conformations over time. A direct comparison of atomic coordinates is not sufficient, since a protein may translate or rotate within the simulation box without undergoing any internal structural change. The structures are therefore first superimposed to remove differences in their overall translation and rotation. Any remaining differences in atomic positions can then be interpreted as conformational changes rather than motion of the protein as a whole.

As introduced in Section 2.4, the position of the full molecular system at time t is represented by $\mathbf{q}(t) \in \mathbb{R}^{3N}$, where N is the total number of atoms. For structural comparison, we restrict our attention to the protein C_α atoms, which provide a compact representation of the protein backbone without including side-chain fluctuations. Let $\mathcal{S}$ denote the set of C_α atoms used for a particular alignment, with $N_{\mathcal{S}} = |\mathcal{S}|$. For each atom $i \in \mathcal{S}$, $\mathbf{q}_i(t)$ denotes its position at time t and $\mathbf{q}_i^{\mathrm{ref}}$ its position in the reference conformation. The reference used throughout is the initial energetically minimised structure.

The current and reference structures are optimally superimposed by allowing a rigid translation and rotation. The optimal transformation at time t is defined by

$$(\mathbf{R}_{\mathcal{S}}^*(t), \mathbf{a}_{\mathcal{S}}^*(t)) = \underset{\mathbf{R}\in SO(3),\, \mathbf{a}\in\mathbb{R}^3}{\operatorname{argmin}} \sum_{i\in\mathcal{S}} \left\| \mathbf{R}\mathbf{q}_i(t) + \mathbf{a} - \mathbf{q}_i^{\mathrm{ref}} \right\|^2, \tag{3.1}$$

where $SO(3)$ denotes the group of three-dimensional rotation matrices, of which $\mathbf{R}$ is an element, and $\mathbf{a}$ is a translation vector. The optimal rotation is obtained using the Kabsch algorithm [37], which solves the corresponding least-squares fitting problem.

The structural alignment described above is used to examine how different parts of the protein move relative to the reference structure. Equation (3.1) is evaluated for the final configuration of each simulation, and the displacement of each C_α atom is calculated after alignment. Two alignment procedures are then considered. In the first, the whole protein is aligned using a single rigid transformation, with $\mathcal{S}$ containing all protein C_α atoms. In the second, the four subdomains IA, IB, IIA and IIB are aligned independently, with a separate optimal transformation calculated for each subdomain.

The whole-protein alignment retains relative motion between the subdomains. This is because a single transformation is applied to the entire protein, so differences in the positions of individual subdomains remain after alignment. By contrast, independent subdomain alignment removes the rigid-body motion of each subdomain and isolates changes arising from internal deformation. The resulting per-residue displacements are shown in Figure 3.1.

Figure 3.1 shows a clear difference between the whole-protein and independently aligned subdomain displacements. Across much of the sequence, the whole-protein displacement is substantially larger. This indicates that much of the observed change arises from rigid-body motion of the subdomains relative to one another. Where both displacements remain small, comparatively little structural change is observed. By contrast, regions with large displacements after independent subdomain alignment indicate substantial internal deformation. These regions are highlighted as flexible regions in Figure 3.1.

The flexible regions comprise the N-terminus, hydrophobic loop and C-terminus. Each lies on the exterior of the protein and corresponds to a flexible structural element highlighted in Figures 2.2 and 2.3b. Their large local fluctuations must be considered carefully, since they can contribute disproportionately to measures of structural change without reflecting the collective motion of the surrounding subdomain. This suggests that these regions should be treated separately, or potentially excluded, when constructing a low-dimensional description of the protein dynamics. We examine this further using root mean square deviation (RMSD) over the full trajectories.

RMSD is a commonly used measure in MD for tracking structural change over time. It extends the structural comparison described above by comparing each configuration

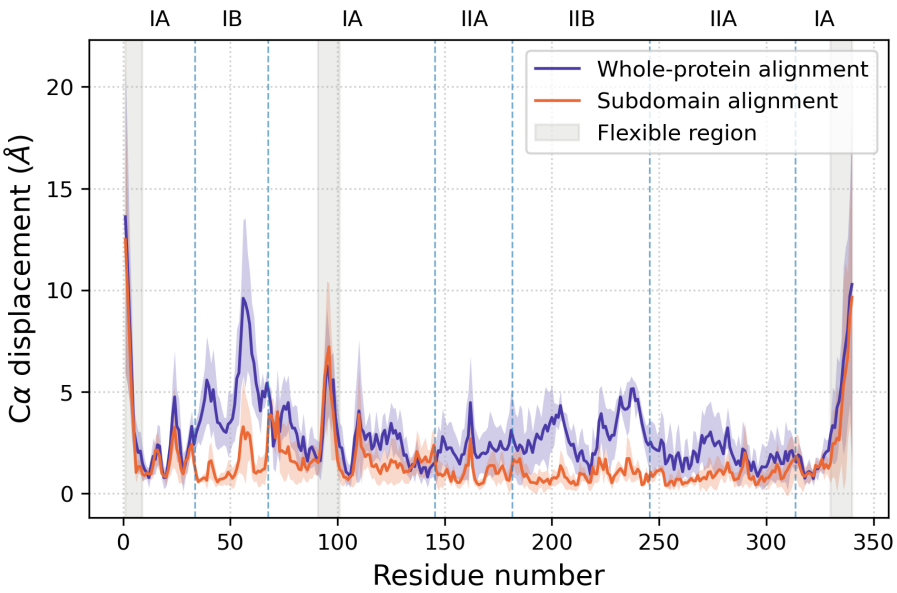


Figure 3.1: Mean final-frame C_α displacement for ATP-bound monomers across five independent simulations. The final structure from each trajectory was compared with the corresponding minimised reference structure using either a single whole-protein alignment or independent alignment of the four structural subdomains. Solid lines show the mean displacement at each residue, while the shaded envelopes indicate the variation between replicas. Dashed vertical lines mark the boundaries between subdomains IA, IB, IIA and IIB. Grey shaded regions indicate the flexible N-terminus (residues 1–9), hydrophobic loop (residues 91–101), and C-terminus (residues 330–340).

in the trajectory with the same reference structure. At each time point, the current structure is first aligned to the reference using Equation (3.1), and the remaining C_α displacements are then combined into a single scalar quantity,

$$\mathrm{RMSD}_{\mathcal{S}}(t) = \sqrt{\frac{1}{N_{\mathcal{S}}} \sum_{i \in \mathcal{S}} \left\| \mathbf{R}^*_{\mathcal{S}}(t)\mathbf{q}_i(t) + \mathbf{a}^*_{\mathcal{S}}(t) - \mathbf{q}_i^{\mathrm{ref}} \right\|^2}.$$

Taking $\mathcal{S}$ as the whole protein gives $\mathrm{RMSD}_{\mathrm{whole}}$, while restricting it separately to each subdomain gives $\mathrm{RMSD}_{\mathrm{IA}}$, $\mathrm{RMSD}_{\mathrm{IB}}$, $\mathrm{RMSD}_{\mathrm{IIA}}$ and $\mathrm{RMSD}_{\mathrm{IIB}}$.

Figure 3.2 compares $\mathrm{RMSD}_{\mathrm{whole}}$ with the four independently aligned subdomain RMSDs across the five ATP-bound monomer replicas, before and after removal of the flexible regions. In panel (a), where all residues are included, $\mathrm{RMSD}_{\mathrm{IA}}$ is relatively large and comparable to $\mathrm{RMSD}_{\mathrm{whole}}$. This is strongly reduced in panel (b) after removal of the N-terminus, hydrophobic loop and C-terminus. Since all three flexible regions belong to IA, their removal substantially reduces $\mathrm{RMSD}_{\mathrm{IA}}$, bringing it closer

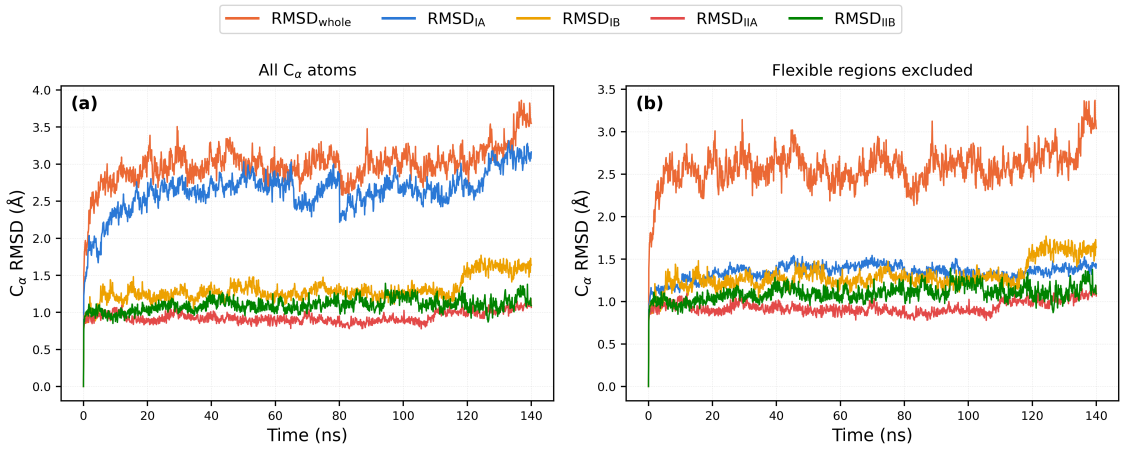


Figure 3.2: Comparison of mean C_α RMSD over the full ATP-bound monomer trajectories, averaged across five independent simulations at each time point. $RMSD_{whole}$ and the independently aligned subdomain quantities $RMSD_{IA}$, $RMSD_{IB}$, $RMSD_{IIA}$ and $RMSD_{IIB}$ are shown. (a) RMSD calculated using all C_α atoms. (b) RMSD calculated after excluding the flexible N-terminus, hydrophobic loop and C-terminus.

to the values of the other subdomains. $RMSD_{whole}$ also decreases, showing that these flexible regions contribute noticeably to the overall structural deviation.

More importantly, after removal of the flexible regions, panel (b) of Figure 3.2 shows that the RMSD of each independently aligned subdomain remains substantially below $RMSD_{whole}$ for most of the trajectory. In other words, motion between the subdomains is greater than deformation within them. This is precisely the type of separation we wanted for a low-dimensional description based on relative subdomain motion. We therefore use the rigid cores of IA, IB, IIA and IIB in the subsequent analysis, excluding the N-terminus, hydrophobic loop and C-terminus. This analysis supports the conventional four-subdomain rigid-core description, with the precise subdomain boundaries examined further in the next section.

3.1.2 Dynamics support the four-subdomain partition

The results of the previous section support a low-dimensional description of $MreB^{Gs}$ based on relative motion between four approximately rigid subdomains. However, this does not establish whether the structurally defined IA, IB, IIA and IIB boundaries provide an optimal, or even particularly good, four-site partition of the trajectory. Principal component analysis (PCA) is widely used to identify collective motions in MD trajectories, and Ng et al. [20] used it to support the conventional four-subdomain

description of MreB^{Tm}. PCA can identify the dominant directions of motion, but it does not provide a direct criterion for comparing alternative four-site partitions. Furthermore, over longer trajectories, the collective motion may also be distributed across several principal components, making the leading modes alone less informative for assessing a proposed partition.

A natural way to formalise this low-dimensional description is through coarse-graining, in which groups of atoms are represented by a smaller number of effective sites. Here, we use the essential-dynamics coarse-graining (ED-CG) method introduced by Zhang et al. [38], which combines coarse-graining with the collective motions identified by PCA. Unlike approaches based primarily on the structural or topological properties of a single conformation [39, 40], ED-CG derives the mapping directly from the simulated dynamics. It therefore provides a natural way to test whether the structurally defined MreB subdomain boundaries are dynamically well motivated.

ED-CG is therefore fundamentally built from PCA. As in the previous section, only the C_α atoms are considered, with overall translation and rotation removed by structural alignment. PCA gives a set of modes $\boldsymbol{\psi}_q$ with corresponding eigenvalues λ_q, where larger λ_q indicate stronger modes of motion. For the $N_{\mathcal{S}}$ C_α atoms included in the analysis, the full coordinate space has dimension $3N_{\mathcal{S}}$, but much of the large-scale collective motion can often be described by a much smaller number of dominant modes, $n_{\mathrm{ED}} \ll 3N_{\mathcal{S}}$ [38]. These largest-eigenvalue modes are referred to as the essential modes, and together span the essential subspace

$$\mathcal{E} = \mathrm{span}\{\boldsymbol{\psi}_1, \ldots, \boldsymbol{\psi}_{n_{\mathrm{ED}}}\}.$$

Zhang et al. [38] take $n_{\mathrm{ED}} = 3N_{\mathrm{CG}} - 6$ for a model containing N_{CG} sites, corresponding to its internal degrees of freedom after overall translation and rotation are removed.

Once the essential subspace has been defined, ED-CG measures how similarly atoms assigned to the same coarse-grained site move within this subspace. The displacement of atom i at time t is denoted by $\Delta\mathbf{r}_i(t)$, and its contribution within the essential subspace by $\Delta\mathbf{r}_i^{\mathrm{ED}}(t)$. If another atom j moves in a correlated manner with atom i, then the difference between their displacements within the essential subspace, $\left|\Delta\mathbf{r}_i^{\mathrm{ED}}(t) - \Delta\mathbf{r}_j^{\mathrm{ED}}(t)\right|^2$, should be small. Let $\mathcal{S}_I \subset \mathcal{S}$ denote the set of C_α atoms assigned to coarse-grained site I. Zhang et al. [38] therefore define the residual

$$\chi^2 = \frac{1}{3N_{\mathrm{CG}}} \sum_{I=1}^{N_{\mathrm{CG}}} \sum_{i \in \mathcal{S}_I} \sum_{\substack{j \in \mathcal{S}_I \\ j \geq i}} \left\langle \left| \Delta\mathbf{r}_i^{\mathrm{ED}} - \Delta\mathbf{r}_j^{\mathrm{ED}} \right|^2 \right\rangle, \tag{3.2}$$

where $\langle\cdot\rangle$ denotes an average over the trajectory frames. A lower χ^2 therefore corresponds to a partition in which atoms within each coarse-grained site move more coherently, leaving more of the collective motion to occur between sites. ED-CG then searches over possible site boundaries and selects the partition that minimises this residual [38].

The residual in Eq. (3.2) can be evaluated directly from the PCA output. From the retained modes $\boldsymbol{\psi}_q$ and their eigenvalues λ_q, the covariance associated with the essential subspace is reconstructed as

$$C^{\mathrm{ED}} = \sum_{q=1}^{n_{\mathrm{ED}}} \lambda_q \, \boldsymbol{\psi}_q \boldsymbol{\psi}_q^T .$$

Thus, C^{ED} contains only the fluctuations represented by the retained essential modes. For atoms i and j, the corresponding scalar covariance is

$$C_{ij}^{\mathrm{ED}} = \left\langle \Delta\mathbf{r}_i^{\mathrm{ED}} \cdot \Delta\mathbf{r}_j^{\mathrm{ED}} \right\rangle .$$

The mean-squared relative displacement between atoms i and j can therefore be obtained directly from these covariance terms,

$$\left\langle \left| \Delta\mathbf{r}_i^{\mathrm{ED}} - \Delta\mathbf{r}_j^{\mathrm{ED}} \right|^2 \right\rangle = C_{ii}^{\mathrm{ED}} - 2C_{ij}^{\mathrm{ED}} + C_{jj}^{\mathrm{ED}} .$$

Hence, the residual χ^2 can be calculated directly from the essential covariance matrix.

Following Section 3.1.1, the flexible regions were excluded before constructing the ED-CG mapping, restricting the analysis to the rigid structural core. After removal of residues 91–101, residues 90 and 102 were treated as adjacent in the retained sequence. This is important because the ED-CG search of Zhang et al. [38] assumes that each coarse-grained site consists of a contiguous segment of the residue sequence. The first 40 ns of each trajectory were discarded as equilibration.

For ED-CG to provide a reliable description, the leading principal components should represent similar collective motions across the independent replicas [38]. This agreement was quantified using the root mean square inner product (RMSIP). For two replicas A and B,

$$\mathrm{RMSIP}(A,B) = \left[\frac{1}{n_{\mathrm{ED}}} \sum_{i=1}^{n_{\mathrm{ED}}} \sum_{j=1}^{n_{\mathrm{ED}}} \left(\boldsymbol{\psi}_i^A \cdot \boldsymbol{\psi}_j^B \right)^2 \right]^{1/2} ,$$

which measures the overlap between their n_{ED}-dimensional essential subspaces. The overall pairwise RMSIP was 0.751, indicating that similar dominant motions were

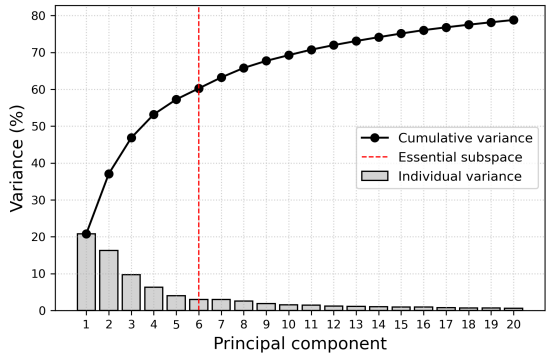
observed across the five replicas. The remaining 100 ns from each trajectory were therefore pooled to improve the statistics of the subsequent PCA and ED-CG analysis.

The search was restricted to four coarse-grained sites, giving $n_{\mathrm{ED}} = 3(4) - 6 = 6$ essential modes. A four-site partition is defined by three boundaries, so with 307 possible boundary positions the search contains $\binom{307}{3} \approx 4.8 \times 10^6$ candidate partitions. This search space was small enough to evaluate exhaustively by calculating χ^2 for every possible set of boundaries. Importantly, the structurally defined IA, IB, IIA and IIB partition is not restricted to contiguous sequence segments and therefore does not belong to this search space. Its χ^2 value was consequently evaluated separately, allowing the conventional, structurally defined partition to be compared with the optimal contiguous four-site ED-CG partition.

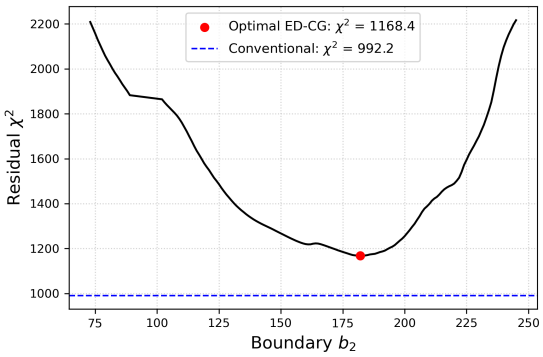
Figure 3.3a shows that the conformational variance is distributed across several principal components. The first six modes nevertheless account for approximately 60% of the total variance, retaining a substantial fraction of the collective motion within the ED-CG essential subspace. The removal of the large initial relaxation reduces the dominance of the leading modes, leaving the more subtle equilibrium fluctuations distributed across several principal components. That these fluctuations can still be represented substantially by only six modes supports the use of a low-dimensional ED-CG description.

Figures 3.3b and 3.3c test whether the structurally determined IA, IB, IIA and IIB decomposition is also a sensible dynamical partition. The minimum residual obtained within the contiguous four-site ED-CG search was $\chi^2 = 1168.4$, whereas the structurally determined non-contiguous partition gave a lower residual of $\chi^2 = 992.2$ (Figure 3.3b). Therefore, no contiguous four-site mapping represented the essential motions better than the structurally determined subdomain decomposition. The optimal ED-CG boundaries shown in Figure 3.3c also lie close to several structurally determined subdomain boundaries, indicating that the exhaustive search independently identifies similar dynamical divisions in the protein.

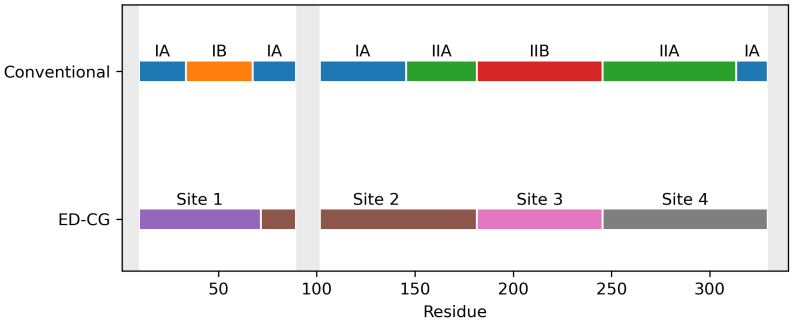
A limitation of ED-CG is that each coarse-grained site must contain residues that are contiguous in sequence [38]. This is restrictive here because the structurally determined subdomains contain regions that are separated in sequence but adjacent in the folded structure. More recent approaches, such as K-means clustering coarse-graining, remove this restriction by allowing non-contiguous sites [41]. Despite this, the close agreement between the ED-CG and conventional boundaries, together with the lower residual of the conventional partition, supports the use of the established



(a) PCA variance



(b) ED-CG search for optimal boundary



(c) Four-site mapping comparison

Figure 3.3: ED-CG analysis of the pooled ATP-bound monomer trajectories using the rigid structural core. (a) Individual and cumulative PCA variance, with the first six modes defining the four-site ED-CG essential subspace. (b) ED-CG search for an optimal boundary, shown as a one-dimensional scan in which b_1 and b_3 are fixed at their optimal positions while b_2 is varied. (c) Comparison of the conventional four-subdomain decomposition with the optimal contiguous ED-CG mapping. Grey regions indicate excluded residues.

four-subdomain description. Combined with the rigidity analysis of Section 3.1.1, this motivates treating IA, IB, IIA and IIB as four effective dynamical units, an idea that will be used again in the coarse-grained model developed later in this report. The next section introduces a simple way of tracking the relative motion between these four sites throughout the simulations.

3.1.3 Opening and dihedral angles as collective variables

The previous sections showed that the dominant conformational motions of MreB^{Gs} can be described largely through the relative motion of four approximately rigid subdomains. This motivates describing their arrangement using a small number of

physically interpretable collective variables. Following previous studies of actin-family proteins, three angles are considered: two internal opening angles, ω_1 and ω_2, and a dihedral angle, ϕ [16, 19]. The opening angles measure hinge-like separation around the nucleotide-binding pocket and therefore describe how open or closed the pocket is. The dihedral angle describes the out-of-plane deformation of the four subdomains, with smaller values corresponding to a flatter conformation. These angles should therefore be regarded as physically motivated collective variables rather than a complete description of the protein structure.

For each trajectory frame, each subdomain was replaced by its mass-weighted centre of mass,

$$\mathbf{r}_S = \frac{\sum_{i \in S} m_i \mathbf{r}_i}{\sum_{i \in S} m_i},$$

where $S \in \{\mathrm{IA}, \mathrm{IB}, \mathrm{IIA}, \mathrm{IIB}\}$ and all atoms in each rigid subdomain are included in the calculation. The two internal opening angles were defined as

$$\omega_1 = \angle(\mathbf{r}_{\mathrm{IIB}}, \mathbf{r}_{\mathrm{IIA}}, \mathbf{r}_{\mathrm{IA}}), \qquad \omega_2 = \angle(\mathbf{r}_{\mathrm{IB}}, \mathbf{r}_{\mathrm{IA}}, \mathbf{r}_{\mathrm{IIA}}).$$

Following previous MreB studies, we define the overall opening angle as the mean of these two internal angles [20],

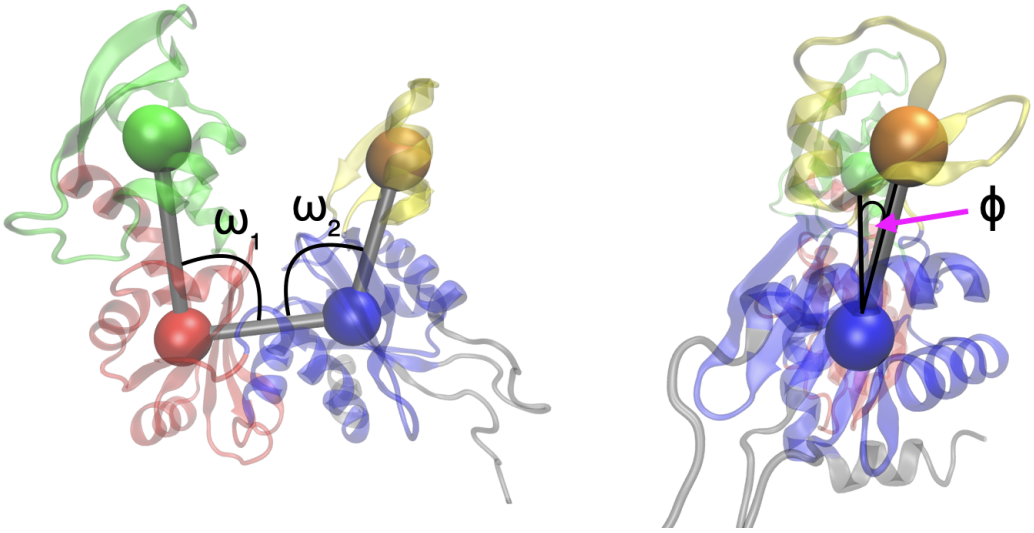
$$\bar{\omega} = \frac{\omega_1 + \omega_2}{2}.$$

This reduces the number of collective variables considered while retaining a measure of the overall opening or closure of the nucleotide-binding pocket.

The dihedral angle ϕ was calculated from the ordered subdomain centres of mass $\mathbf{r}_{\mathrm{IB}} - \mathbf{r}_{\mathrm{IA}} - \mathbf{r}_{\mathrm{IIA}} - \mathbf{r}_{\mathrm{IIB}}$. It is defined as the angle between the planes containing $(\mathbf{r}_{\mathrm{IB}}, \mathbf{r}_{\mathrm{IA}}, \mathbf{r}_{\mathrm{IIA}})$ and $(\mathbf{r}_{\mathrm{IA}}, \mathbf{r}_{\mathrm{IIA}}, \mathbf{r}_{\mathrm{IIB}})$. Letting $\mathbf{n}_1$ and $\mathbf{n}_2$ denote normal vectors to these two planes, respectively, the magnitude of the dihedral angle is given by

$$\phi = \cos^{-1}\left(\frac{\mathbf{n}_1 \cdot \mathbf{n}_2}{\|\mathbf{n}_1\| \, \|\mathbf{n}_2\|}\right).$$

Schematics of the opening and dihedral angles are shown in Figure 3.4.



(a) Internal opening angles ω_1 and ω_2.

(b) Dihedral angle ϕ.

Figure 3.4: Internal opening and dihedral angles describing the relative arrangement of the four MreB^{Gs} subdomains. Coloured spheres indicate the subdomain centres of mass.

3.2 Conformational Variability Across Simulations

We now examine whether the three conditions considered in this work: nucleotide state, polymerisation and membrane association, produce reproducible differences in the conformation of MreB^{Gs}. Based on Section 3.1, these comparisons are made primarily using the opening angle $\bar{\omega}$ and the dihedral angle ϕ, which capture the dominant large-scale motions of the protein. However, any apparent differences between conditions must be interpreted alongside variability between independent replicas and the possibility of incomplete sampling of metastable conformational states. This section therefore asks what conclusions the simulations actually support given these limitations.

3.2.1 Finite trajectories require careful interpretation

Previous MreB MD studies also used changes in opening and dihedral angles to characterise conformational behaviour [19–21]. In these studies, simulations were run until these observables were considered "equilibrated", based on whether their distributions over a chosen time interval could be fitted by Gaussian distributions. These conclusions were generally based on only one or two independent replicas, which did not always sample the same conformational ranges or produce the same mean val-

ues. Later microsecond-scale simulations further showed that apparently stable MreB conformations can persist for long periods before transitioning to other quasi-stable states [22]. Together, these observations motivate a more careful assessment of finite sampling and replica variability.

This behaviour can be understood in terms of the energy landscape explored by an MD trajectory. As discussed in Section 2.4, atomic motion is governed by the potential energy $U(\mathbf{q})$ together with thermal fluctuations from the Langevin dynamics. The resulting energy landscape is generally rugged, containing many low-energy regions separated by barriers [42, 43]. These regions may correspond to distinct, physically meaningful conformations, and a trajectory may remain within one region for a long period before crossing into another. A finite simulation may therefore sample only part of the accessible landscape, so an observable can appear stable without necessarily reflecting the full conformational behaviour of the protein.

Multiple independent replicas help address this limitation. Simulations started from the same coordinates and run with identical parameters can follow different paths because of their random initial velocities and the stochastic forcing in the Langevin dynamics. Successive frames within a single trajectory are also correlated, so a large number of saved configurations does not correspond to the same number of independent samples. Separate replicas therefore provide genuinely independent trajectories and increase the amount of independent information available for analysis.

Independent replicas also reveal whether an observed trend is reproducible across different trajectories. This is particularly important when transitions between conformations are rare, since extending a single trajectory does not guarantee that such transitions will be observed. Knapp et al. [44] showed that conclusions from single trajectories can be non-reproducible, whereas multiple independent replicas generally provide more reliable estimates. The five replicas performed for each simulated condition in the present work therefore provide a stronger basis for assessing robustness than the one or two replicas typically used in the earlier MreB studies [19–21]. These considerations are kept in mind below when examining the conformational behaviour and reproducibility of the $\mathrm{MreB}^{\mathrm{Gs}}$ trajectories.

3.2.2 Replica variability persists beyond initial relaxation

The five ATP-bound monomer replicas are used here as a representative example of the sampling behaviour discussed above. Figure 3.5 shows the opening and dihedral

angles for these replicas, illustrating how independent runs under the same conditions can follow different conformational histories. Both angles show slow drift, long-lived plateaus and occasional abrupt changes, some occurring well beyond the initial relaxation period. For example, the dihedral angle of replica 1 remains relatively stable for much of the simulation before increasing sharply near the end of the trajectory. An observable can therefore appear stable for an extended period before moving into a different conformational range.

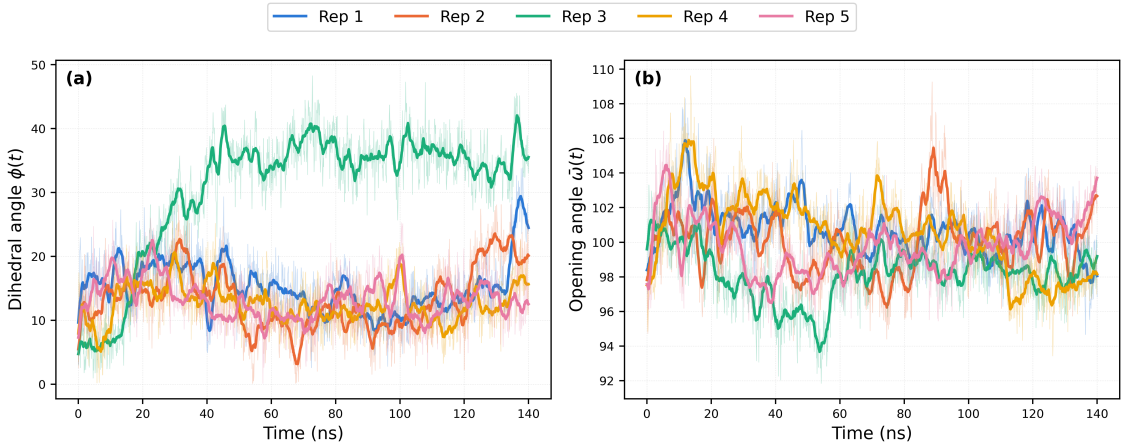


Figure 3.5: Observable trajectories of the five ATP-bound MreB^{Gs} monomer replicas. Smoothed trajectories are shown for (a) the dihedral angle ϕ and (b) the opening angle $\bar{\omega}$, using a 20-frame moving average. Faint lines show the corresponding unsmoothed trajectories.

The variability can be separated into within-replica and between-replica contributions. Within a single trajectory, an observable may fluctuate, drift or transition between long-lived conformational states. Between-replica variability instead reflects differences in the conformational regions sampled by independent trajectories. This is particularly clear for the dihedral angle of replica 3, which occupies a substantially higher-angle region than most other replicas for much of the simulation. The two observables also evolve differently, so apparent stability in one coordinate does not imply similar behaviour in another.

Taken together, these effects make comparisons based on single late-time mean values difficult. An apparently stable or approximately Gaussian interval does not necessarily imply that all relevant conformational states have been sampled. The equilibration criterion used in earlier MreB studies is therefore not well suited to the present simulations, where slow changes and replica-to-replica differences persist at late times.

Conclusions must instead be based on behaviour across multiple replicas and interpreted with the limitations of finite sampling in mind.

Before comparing the different simulation conditions, an initial relaxation period was excluded from each trajectory. Since the starting structure may not correspond to the preferred conformation of a given system, the early trajectory can be dominated by structural relaxation rather than representative sampling. Previous MreB studies treated a similar interval as a "burn-in" preceding equilibration, however, here we refer only to initial structural relaxation rather than assigning a precise equilibration time. Because this relaxation can overlap with slower drift and transitions between long-lived conformations, the cutoff is necessarily approximate. Based on the ATP-bound monomer trajectories, the first 40 ns were excluded, beyond which the within-replica variability changed relatively little with further increases in the cutoff. This is comparable to the initial periods excluded in earlier MreB studies, and the same 40 ns cutoff was applied consistently to all simulated conditions.

3.2.3 Monomer variability obscures nucleotide differences

We first asked whether nucleotide state affects the conformation of $MreB^{Gs}$ monomers. Previous MD studies of Gram-negative MreB homologues suggested nucleotide-dependent differences in opening and dihedral angles. For $MreB^{Tm}$, ATP-bound monomers were reported to adopt larger opening and dihedral angles than the ADP-bound form [19, 20]. $MreB^{Cc}$ showed the same reported trend in opening angle, but the opposite dihedral response, with ADP producing the larger angle [21]. Together, these studies suggested that nucleotide influences MreB conformation, although the response may differ between homologues. Here, using multiple independent replicas, we instead ask whether any ATP–ADP difference is reproducible across trajectories.

Figure 3.6 compares the distributions of $\bar{\omega}$ and ϕ for ATP- and ADP-bound monomers across replicas after the initial relaxation period. Kernel density estimates (KDEs) provide smooth representations of the sampled distributions without imposing the single Gaussian form used in previous MreB studies. Thick curves show the distributions pooled across all five replicas, while faint dashed curves show individual replicas. Both nucleotide states sample broad, strongly overlapping ranges of opening and dihedral angles, with several apparent peaks and extended tails. These distributions should therefore be interpreted as finite-sampling estimates rather than fully converged probability distributions.

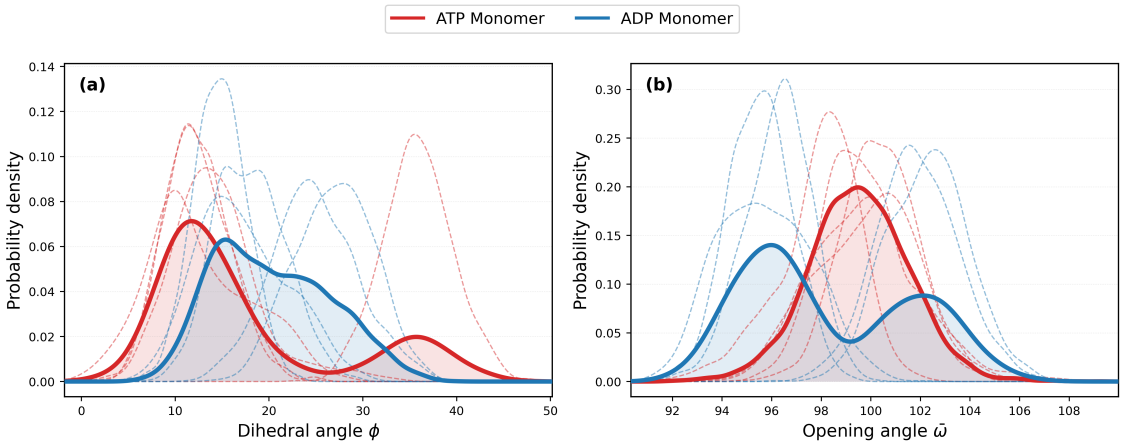


Figure 3.6: Conformational distributions of ATP- and ADP-bound MreB^{Gs} monomers. Kernel density estimates are shown for (a) the dihedral angle ϕ and (b) the opening angle $\bar{\omega}$, using frames after the initial 40 ns relaxation period. Thick solid curves show the distributions pooled across all five replicas for each nucleotide state, while faint dashed curves show the corresponding individual-replica distributions.

The individual replica curves show that each trajectory often samples a relatively narrow conformational range, but that these ranges differ substantially between replicas. This between-replica variability is larger than the apparent ATP–ADP difference: for example, the mean dihedral angles of the ATP replicas span approximately 24°, whereas the overall ATP and ADP means differ by only about 3°. The multiple peaks in the pooled KDEs therefore arise largely because different replicas occupy different conformational regions. With only one or two trajectories, the apparent nucleotide-dependent behaviour could therefore depend strongly on which replicas were sampled. We therefore do not consider the small differences between the pooled ATP and ADP distributions sufficient evidence for a reproducible shift in monomer opening or dihedral angle.

The absence of a reproducible ATP–ADP shift should not be interpreted simply as a negative result. Rather, the broad range of opening and dihedral angles sampled across the replicas suggests that MreB^{Gs} monomers are conformationally flexible, rather than occupying a single well-defined "open", "closed", "flat" or "twisted" state. This is consistent with more recent simulations showing that MreB can occupy several long-lived conformational states [22]. Our results suggest that a similar picture may also apply to isolated MreB^{Gs} monomers, which sample a wide range of opening and dihedral angles rather than a single well-defined conformation. Nucleotide state may still influence which conformations are favoured, but the present trajectories do

not sample this landscape sufficiently to attribute the small ATP–ADP differences confidently to nucleotide state.

3.2.4 Polymerisation and membrane binding limit flexibility

We next examined how polymerisation and membrane association influence MreB^{Gs} conformation. Previous simulations of Gram-negative MreB homologues reported polymerisation-dependent changes in opening and dihedral angles, particularly closure and flattening [19, 21]. Membrane association is also of particular interest for MreB^{Gs}, since filament formation has been observed predominantly on lipid surfaces [3]. We therefore compared solution monomers, solution dimers and membrane-bound dimers to assess how these structural environments affect conformation.

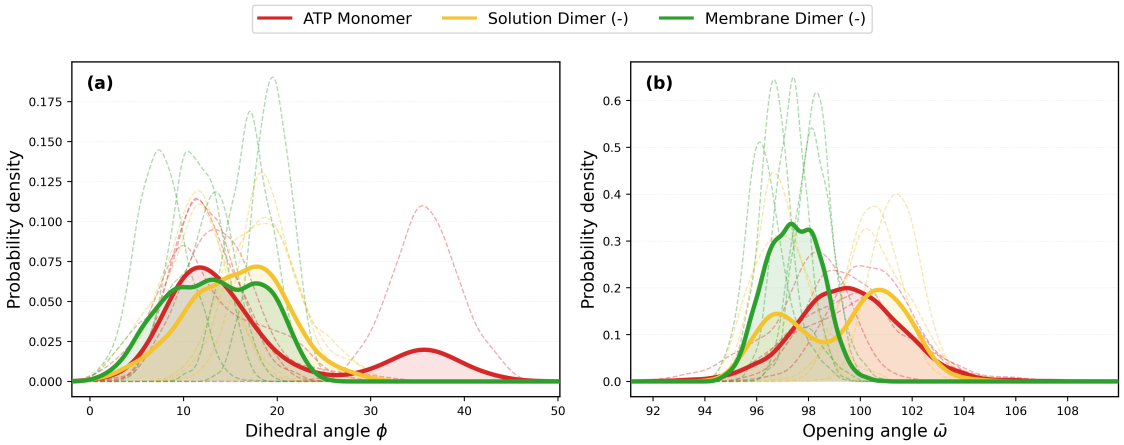


Figure 3.7: Conformational distributions of ATP-bound MreB^{Gs} in the monomer, solution dimer and membrane-bound dimer states. For both dimer systems, distributions are shown for the (−) subunit. Kernel density estimates are shown for (a) the dihedral angle ϕ and (b) the mean opening angle $\bar{\omega}$, using frames after the initial 40 ns relaxation period. Thick solid curves show the distributions pooled across all replicas for each condition, while faint dashed curves show the corresponding individual-replica distributions.

Figure 3.7 shows that membrane association produces the clearest reduction in conformational variability. For both dimer systems, the distributions shown are for the (−) subunit. The effect is strongest for $\bar{\omega}$: the membrane-bound replicas occupy a much narrower and more consistent range than either the monomer or solution dimer. The difference in ϕ is less pronounced. Both dimer systems avoid the highly twisted conformations sampled by some monomer replicas, while the membrane-bound distribution is slightly narrower and shifted towards lower ϕ than the solution dimer. Overall,

conformational sampling becomes progressively more restricted from the monomer to the solution dimer and then to the membrane-bound dimer.

These results suggest that conformational variability decreases as additional structural constraints are introduced. The free monomer has no inter-subunit contacts, whereas polymerisation introduces the longitudinal interface between neighbouring subunits shown in Figure 2.3b. Membrane association adds a further constraint through interactions between the membrane-facing regions of $MreB^{Gs}$ and the lipid surface shown in Figure 2.3c. The progressive narrowing of the conformational distributions is therefore consistent with the increasing constraints across these systems. This effect is much clearer than the nucleotide-dependent differences observed for isolated monomers, where between-replica variability dominated the comparison. ATP- and ADP-bound solution dimers likewise showed substantial overlap and no reproducible nucleotide-dependent shift across replicas (Figure B.1, Appendix).

Having established a strong membrane-associated reduction in conformational flexibility, we next asked whether membrane binding alone is sufficient to produce this effect, or whether polymerisation also contributes. To address this, we compared the (+) and (−) subunits within the membrane-bound dimers. Although both subunits contact the membrane and form part of the dimer, the (−) subunit is more directly constrained by the longitudinal interface. Differences between the two subunits can therefore provide an indication of the additional conformational restriction associated with polymerisation.

Figure 3.8 shows that the reduced conformational flexibility of the membrane-bound dimer is not shared equally by its two subunits. The clearest difference is again seen in $\bar{\omega}$. Although the pooled (+) and (−) distributions have similar means, the (−) replicas occupy a much narrower and more consistent range, whereas the (+) subunit shows greater within- and between-replica variability. The difference in ϕ is weaker, but the (−) subunit again samples a somewhat narrower range and avoids some of the larger values reached by the (+) subunit.

This asymmetry suggests that membrane association alone does not fully explain the restricted conformational sampling of the (−) subunit. Instead, the different structural constraints experienced by the two subunits appear to contribute to their different flexibility. This is consistent with the importance of membrane association for $MreB^{Gs}$ filament formation [3] and suggests that membrane association may help stabilise the polymerised state. The next section examines this more directly by analysing the stability of the longitudinal interface.

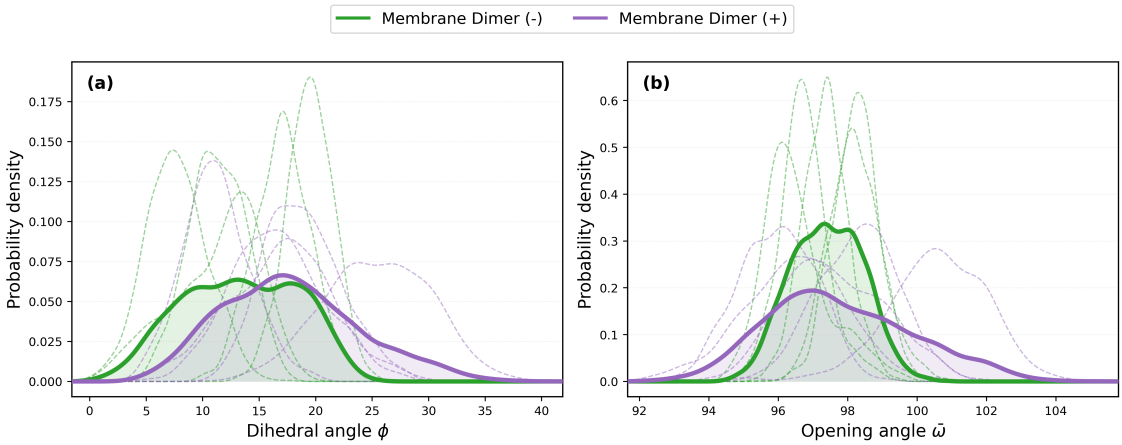


Figure 3.8: Conformational distributions of the (−) and (+) subunits of ATP-bound membrane-associated MreB^{Gs} dimers. Kernel density estimates are shown for (a) the dihedral angle ϕ and (b) the mean opening angle $\bar{\omega}$, using frames after the initial 40 ns relaxation period. Thick solid curves show the distributions pooled across all replicas for each subunit, while faint dashed curves show the corresponding individual-replica distributions.

3.3 Membrane Effects on Dimer Stability

The results above show that the membrane-bound dimer samples a substantially more restricted conformational ensemble than either the monomer or solution dimer systems. This raises the question of what physical interactions produce this additional constraint. In this section, we therefore examine the principal effects of membrane association on the polymerised MreB^{Gs} dimer, with the aim of identifying how the lipid environment alters and stabilises its conformation.

3.3.1 Membrane binding stabilises the longitudinal interface

One important feature not yet considered is whether the longitudinal interface of the dimer itself remains intact. We therefore asked whether membrane association alters the way in which the two subunits remain connected. As shown in Figure 2.3b, the longitudinal interface is formed primarily through contacts between subdomains IB(−)–IA(+) and IIB(−)–IIA(+). As a first indicator of interface behaviour, we examined these inter-subunit contacts at the coarse-grained subdomain level.

Figure 3.9 reveals a clear difference between the two regions of the longitudinal interface. The most striking behaviour occurs for the IB(−)–IA(+) separation (Figure 3.9a). In the membrane-bound simulations, this distance remains close to its initial value of approximately 30 Å across all replicas. In solution, the behaviour

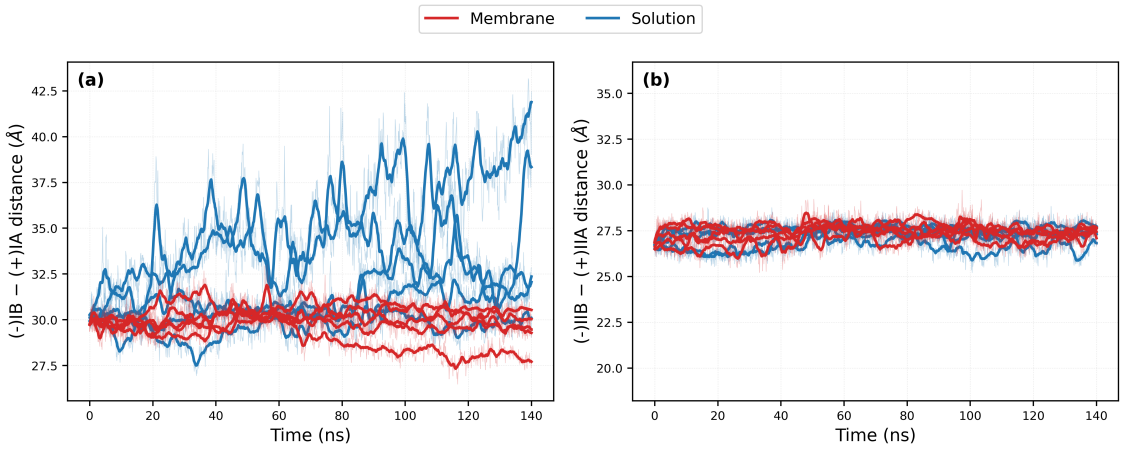


Figure 3.9: Inter-subunit subdomain distances for the ATP-bound MreB^{Gs} dimer in solution and at the membrane. Smoothed trajectories, using the 20-frame moving average, are shown for (a) the distance between the centres of mass of subdomains IB(−) and IA(+) and (b) the distance between subdomains IIB(−) and IIA(+). Faint lines show the corresponding unsmoothed trajectories. Membrane-bound replicas are shown in red and solution replicas in blue. Both panels span approximately 15 Å on the vertical axis to allow a fair visual comparison of the fluctuations.

is more variable: several replicas progressively move to much larger separations, in some cases exceeding 35–40 Å, whereas others remain closer to the membrane-bound range. The overall trend nevertheless indicates that the IB(−)–IA(+) region is less consistently maintained in solution. By contrast, the IIB(−)–IIA(+) distance remains tightly clustered at approximately 27 Å in both environments (Figure 3.9b). The membrane therefore appears to stabilise specifically the IB(−)–IA(+) side of the longitudinal interface, rather than uniformly constraining the entire dimer.

This distinction is particularly interesting for MreB^{Gs}, because the IB(−)–IA(+) contact lies on the membrane-facing side of the dimer (Figure 2.3c). As described in Section 2.3.3, this contact is a distinctive feature of the relatively open MreB^{Gs} interface, where IB interacts with IA of the neighbouring subunit rather than IIA, as in more closed homologues such as MreB^{Tm} [3]. The present simulations therefore suggest that this distinctive inter-subunit contact may be particularly dependent on membrane association for its stability. We next examine which membrane interactions may contribute to this stabilisation.

3.3.2 Membrane association persists across simulations

$MreB^{Gs}$ associates with the membrane through two membrane-facing regions: the N-terminus and the hydrophobic loop. To examine how these regions interact with the membrane, we first defined a frame-dependent reference for the upper membrane surface. At each frame, the mean z-position of the phosphorus atoms in the upper membrane layer was calculated as

$$z_{\mathrm{surf}}(t) = \frac{1}{N_{\mathrm{upper}}} \sum_{i=1}^{N_{\mathrm{upper}}} z_{P,i}(t). \tag{3.3}$$

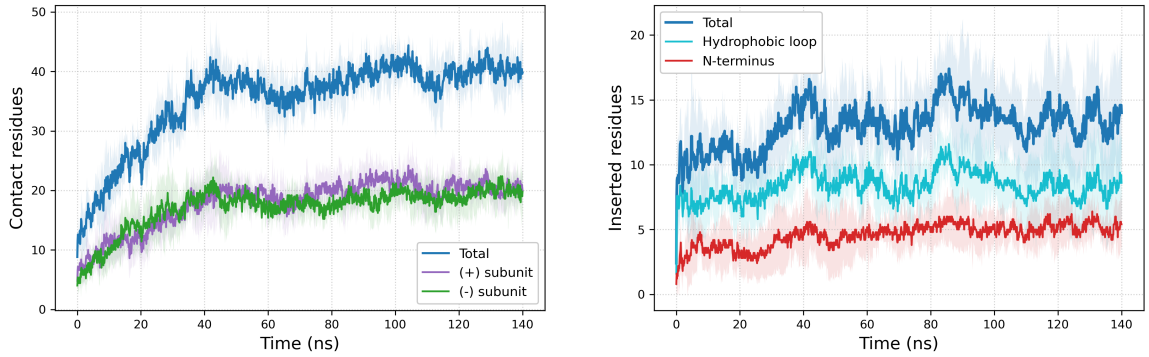
where $z_{P,i}(t)$ is the z-coordinate of phosphorus atom i in the upper membrane layer at time t, and N_{upper} is the total number of phosphorus atoms in that layer. The membrane was then approximated as a flat plane at $z = z_{\mathrm{surf}}(t)$. This plane was used as the reference for all subsequent height and insertion measurements.

There were two main aspects of membrane association that we wanted to quantify: insertion of the binding regions and the overall protein–membrane contact. To measure insertion, we tracked the N-terminal and hydrophobic-loop regions relative to the membrane surface. For each residue in these regions, the mean z-position of its heavy atoms were calculated, and the residue was counted as inserted when this value fell below $z_{\mathrm{surf}}(t)$.

Overall protein–membrane contact was defined using a 5 Å cutoff, chosen to identify residues close enough to participate in short-range interactions with the membrane. A residue was classified as contacting the membrane if at least one heavy atom lay within 5 Å of a phosphorus atom in the upper membrane layer. During system construction, the protein was initially positioned above the membrane, after which membrane association was allowed to develop during equilibration and the subsequent simulation.

Figure 3.10a shows that the overall protein–membrane contact develops progressively during the first ∼40 ns. The mean number of contacting residues rises from approximately 10 to around 35–40, after which it fluctuates around a broadly stable plateau. The (+) and (−) subunits make similar contributions throughout, with neither showing a consistently greater degree of membrane contact. The dimer therefore develops from relatively limited initial contact to a persistent membrane-associated state.

Figure 3.10b shows that the number of inserted residues rises rapidly at the beginning of the trajectories and then remains substantial throughout the simulations. Both



(a) Protein–membrane close contacts

(b) Membrane-region insertion

Figure 3.10: Membrane association of the ATP-bound MreB^{Gs} dimer. (a) Number of protein residues within 5 Å of an upper-leaflet phosphate atom, shown for the complete dimer and separately for the (+) and (−) subunits. (b) Number of membrane-binding-region residues below the upper-leaflet phosphate surface, separated into hydrophobic-loop and N-terminal contributions. Solid lines show the mean across five replicas and shading indicates ±1 standard deviation.

the hydrophobic loop and N-terminal region contribute to this behaviour, with the hydrophobic loop accounting for the larger number of inserted residues. The wider standard-deviation bands show that the precise extent of insertion varies between replicas. Despite this variability, the membrane-binding regions remain substantially inserted throughout all five simulations.

Taken together, these results suggest how the membrane may provide an additional stabilising constraint on the dimer. The N-terminal and hydrophobic-loop regions remain persistently inserted, anchoring the membrane-facing side of the protein. The plateau in close contacts suggests that the dimer reaches a preferred membrane-associated position rather than continuing to penetrate the bilayer. This may reflect a balance between favourable membrane interactions and steric or short-range repulsive forces at close separation. These interactions may restrict motion of the membrane-facing side and help stabilise the IB(−)–IA(+) region of the longitudinal interface. The specific interactions responsible are not resolved here and are examined more closely in the next section.

3.3.3 Membrane association restricts C-terminal mobility

The C-terminal region provides an obvious candidate for understanding stabilisation of the IB(−)–IA(+) side of the longitudinal interface. As described in Section 2.3.3

and shown in Figure 2.3b, the unusually open conformation of MreB^{Gs} brings subdomain IB into contact with the C-terminal region of IA in the neighbouring subunit [3]. In more closed MreB homologues, IB instead lies closer to subdomain IIA. The separation observed in Section 3.3.1 therefore occurs at this distinctive C-terminal contact.

There are also independent indications that the C-terminus is particularly mobile. In Section 3.1.1, this region was identified as flexible and was therefore excluded from the subsequent rigid-body analysis. Consistent with this, parts of the C-terminus are unresolved in both the ATP- and ADP-bound MreB^{Gs} monomer structures 8AZG and 8AAM [3], suggesting that this region is conformationally flexible. Together, these observations raise the question of whether C-terminal mobility contributes to weakening the IB(−)–IA(+) interface in solution, and whether membrane association stabilises this contact by restricting that motion.

To quantify C-terminal mobility, we measured the displacement of the C-terminal region from its initial position. As in Section 3.1.1, the final structure from each trajectory was first aligned to the corresponding minimised reference structure. The alignment was performed using the stable core of subdomain IA, with the flexible N-terminal and hydrophobic-loop regions, together with the C-terminus itself, excluded from the fitting region. The displacement of each C_α atom in residues 330–340 was then calculated relative to its corresponding position in the reference structure as

$$d_{\mathrm{Cterm}} = \frac{1}{11} \sum_{i=330}^{340} \left\| \mathbf{r}_i^{\mathrm{final}} - \mathbf{r}_i^{\mathrm{ref}} \right\|,$$

where $\mathbf{r}_i^{\mathrm{final}}$ and $\mathbf{r}_i^{\mathrm{ref}}$ denote the aligned final and reference positions of residue i, respectively. The resulting d_{Cterm} was then used to compare C-terminal motion between the different structural environments. A larger value of d_{Cterm} therefore indicates greater movement of the C-terminal region relative to the IA core, whereas a smaller value indicates that the C-terminus remains more closely constrained to its initial position.

To determine how membrane association and longitudinal interface formation affect C-terminal mobility, we compared the (−) and (+) subunits in the ATP-bound solution and membrane-bound dimers. As illustrated in Figure 2.3b, the C-terminus of the (−) subunit does not participate in the IB(−)–IA(+) contact, whereas the C-terminus of the (+) subunit forms part of this interface. The four conditions in Figure 3.11 therefore represent four C-terminal states: solution without interface engagement (yellow), solution with interface engagement (pink), membrane-bound

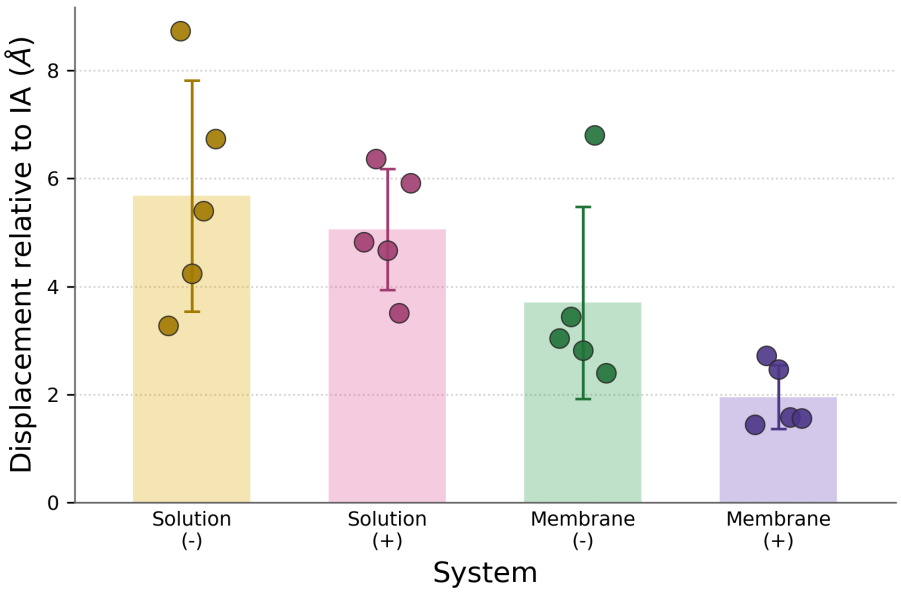


Figure 3.11: C-terminal displacement relative to subdomain IA for the ATP-bound solution and membrane-bound $MreB^{Gs}$ dimers. For each replica, the final structure at 140 ns was aligned to the minimised reference structure using the stable IA core, after which the mean C_α displacement of residues 330–340 was calculated. Bars show the mean across five replicas, individual points show each replica, and error bars indicate ±1 standard deviation.

without interface engagement (green), and membrane-bound with interface engagement (purple).

Figure 3.11 shows that the C-terminus is most strongly stabilised when membrane association and longitudinal interface engagement are both present. The membrane-bound (+) subunit has a mean displacement of approximately 2 Å, approximately threefold lower than the in solution (−) case, and all five replicas remain close to the reference structure. By comparison, the two solution conditions remain centred at much larger displacements of approximately 5–6 Å, indicating that the C-terminus is considerably less constrained in the absence of the membrane.

The comparison between conditions suggests that membrane association contributes more strongly to C-terminal stabilisation than interface engagement alone. In solution, the (+) and (−) subunits show similar displacements, indicating that longitudinal contact formation by itself provides little additional constraint. By contrast, the membrane-bound (−) subunit is already substantially more restricted, while the membrane-bound (+) subunit shows the smallest displacement of all. This suggests that the membrane can constrain the C-terminus independently, with interface en-

gagement providing an additional stabilising effect. As the C-terminus is not itself a membrane-binding region, this influence is likely indirect, possibly arising from steric or repulsive interactions with the membrane surface.

Together, these results provide a plausible explanation for the stability of the IB(−)–IA(+) interface in the membrane-bound simulations. In solution, the C-terminal region remains comparatively mobile even while participating in the longitudinal contact. Membrane association restricts this motion, providing an additional structural constraint that may help maintain the C-terminal contact with IB(−) and prevent the separation observed in solution. More broadly, this supports a stabilising role for the membrane in the polymerised dimer and motivates its explicit inclusion in the coarse-grained model developed in the following chapter.

Chapter 4

Coarse-Grained Model

The all-atom simulations in the previous chapter provide a detailed description of $MreB^{Gs}$ dynamics, but at a substantial computational cost that limits the accessible system size and simulation timescale. Coarse-grained modelling offers a route to retain the most important features of these trajectories while enabling larger assemblies and longer-timescale behaviour to be considered. In this chapter, a coarse-grained model of $MreB^{Gs}$ is developed using information extracted directly from the MD simulations, following the general approach introduced for actin by Chu and Voth [45]. Crucially, this work introduces a bespoke extension of the framework that directly incorporates membrane interaction, a defining feature of MreB that distinguishes it from actin and is particularly important for $MreB^{Gs}$. The aim is to develop and validate this reduced representation as a proof of concept for future modelling of larger MreB assemblies.

4.1 Coarse-Grained Model Formulation

Previous coarse-grained models of MreB have primarily focused on filament-scale behaviour. For example, Shi et al. [21] represented the double filament as a continuous beam to investigate how bending and twist influence filament length and orientation. Such models do not retain the internal structure of individual MreB subunits. Here, a more structurally detailed representation is adopted, following the coarse-grained actin model of Chu and Voth [45], in which each subunit is represented by four sites. This allows the opening and dihedral angles of each subunit, together with changes across the longitudinal interface, to be described explicitly. The four-site representation is also supported by the analysis in Section 3.1, where the $MreB^{Gs}$ subdomains behaved approximately as coherent dynamical units. We first introduce

the original Chu–Voth model before adapting it to membrane-bound MreB^{Gs}.

4.1.1 Previous coarse-grained modelling of actin

Chu and Voth developed their coarse-grained description of actin across two related studies [16,45]. Their earlier work used coarse-grained models of actin monomers and trimers to compare the structural and mechanical effects of ATP and ADP. In later work, the framework was extended to a periodic 13-monomer F-actin filament and then applied to larger assemblies. Conceptually, the model can be viewed as an elastic network of beads connected by effective "springs". Each actin monomer is reduced to four beads corresponding to its structural subdomains (IA, IB, IIA and IIB), with the interactions between them represented by simple effective potentials.

The effective potentials describe both intra- and intermonomer motion. Intramonomer motion is represented by a natural set of six effective internal coordinates (EICs): three bond lengths, two opening angles and one dihedral angle, which describe the relative motion of the four subdomains. Defining the intermonomer EICs is less straightforward, since there is no similarly obvious set of effective coordinates. Chu and Voth therefore represented these more generally using harmonic distances between beads on different monomers. For each pair of four-site monomers, all $4 \times 4 = 16$ possible site–site distances are considered, leading to a large network of candidate interactions across the filament [45]. These should be interpreted as effective mechanical couplings rather than literal physical contacts.

The resulting interactions are described by simple elastic potentials. For EICs such as bond lengths and angles, the energy of the jth EIC is represented by the harmonic potential,

$$E_j = \frac{1}{2} k_j \left(q_j - q_j^0\right)^2, \tag{4.1}$$

where q_j^0 is the equilibrium value and k_j is the corresponding effective stiffness. The harmonic form assumes small fluctuations about a local equilibrium. The dihedral angle is treated using a periodic potential with the same local harmonic behaviour. The k_j values are determined from atomistic fluctuations using the fluctuation-matching procedure described later in Section 4.2, with redundant interactions driven towards zero stiffness.

An important consequence of this coarse-graining framework is the number of degrees of freedom retained in the model. A system of N coarse-grained sites has $3N$ Cartesian degrees of freedom, of which six correspond to translation and rotation of the structure

as a whole. These whole-protein motions are normally discarded, leaving $3N - 6$ internal degrees of freedom described by the effective potentials. This treatment is appropriate for an isolated protein, but becomes less straightforward when the protein interacts with an external surface, such as a membrane.

4.1.2 A membrane-referenced model of MreB

The $3N - 6$ description above assumes that translation and rotation of the coarse-grained structure as a whole carry no energetic significance. For membrane-bound MreB^{Gs}, however, the membrane provides an external spatial reference against which some of these motions become physically meaningful. In the atomistic simulations, the dimer moves towards the membrane before remaining at a relatively stable height above it (Figure 3.10b). This motivates introducing the membrane explicitly into the coarse-grained model rather than automatically discarding these whole-protein motions.

The membrane is represented as a flat reference surface in the xy-plane, with its position defined using $z_{\text{surf}}(t)$ from Section 3.3.2 (Equation 3.3) and shifted so that the surface lies at $z = 0$. Selected coarse-grained sites are then effectively anchored to this surface in the z-direction, with their heights measured relative to the membrane plane. For each anchored site a, the membrane restraint uses the same harmonic form as Equation (4.1),

$$E_{\text{m},a} = \frac{1}{2} k_{\text{m},a} \left(z_a - z_a^0 \right)^2 ,$$

where z_a^0 is the equilibrium height and $k_{\text{m},a}$ is the corresponding effective stiffness. These bead heights form an additional set of membrane-referenced EICs. Hereafter, the term EIC is used collectively for both these coordinates and the protein EICs defined above.

Figure 4.1 shows how the membrane restraint affects the six whole-protein motions of the coarse-grained structure: three translations, T_x, T_y and T_z, and three rotations, R_x, R_y and R_z, along or about the x-, y- and z-axes. Because the membrane potential depends only on height, T_z, R_x and R_y alter the anchor heights and are therefore constrained. These motions correspond to vertical displacement and tilting relative to the membrane. In contrast, T_x, T_y and R_z leave the anchor heights unchanged, corresponding to sliding across the membrane surface or rotation within its plane. These motions remain unconstrained and at zero energy. The number of mechanically relevant degrees of freedom therefore increases from $3N - 6$ to $3N - 3$.

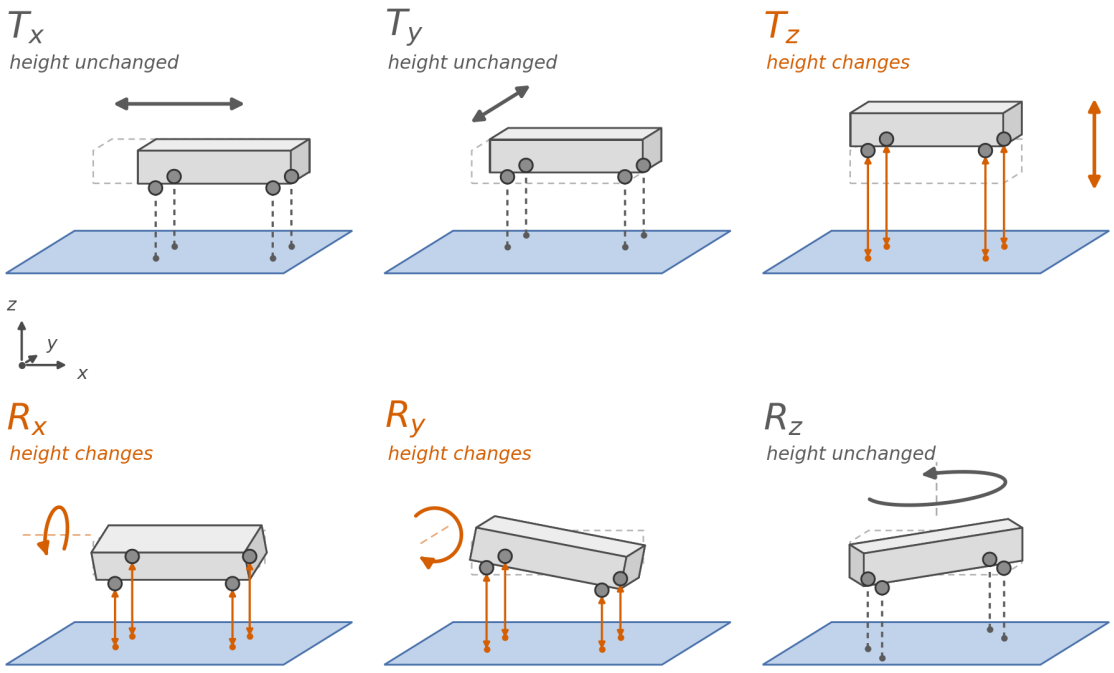


Figure 4.1: Schematic showing the effect of a height-dependent membrane restraint on the six whole-protein translations and rotations. Top row: translations along the x-, y- and z-directions (T_x, T_y and T_z). Bottom row: rotations about the corresponding axes (R_x, R_y and R_z). Motions highlighted in orange (T_z, R_x and R_y) change the anchor heights and are constrained by the membrane restraint, whereas T_x, T_y and R_z leave them unchanged.

With this framework established, we now construct a coarse-grained model of the membrane-bound MreB^{Gs} dimer. This system was chosen because membrane-bound ATP dimer simulations were available, providing the data required to parameterise the model. Following Chu and Voth, and supported by the subdomain analysis in Chapter 3, each monomer is represented by four coarse-grained sites corresponding to the approximately rigid subdomains (IA, IB, IIA and IIB). The resulting eight-bead model includes interactions across the longitudinal interface and with the membrane surface, as illustrated in Figure 4.2.

The eight coarse-grained beads define a Cartesian configuration $\mathbf{x}(t) \in \mathbb{R}^{24}$, with $\mathbf{x}_0$ denoting the reference configuration. Unlike the Chu–Voth model, where only relative bead positions are important, the membrane provides an external reference that makes the bead z-coordinates physically meaningful. Each trajectory frame is therefore re-centred so that the membrane surface lies at $z = 0$, ensuring that bead heights describe vertical motion relative to the membrane plane rather than drift of the membrane or simulation box.

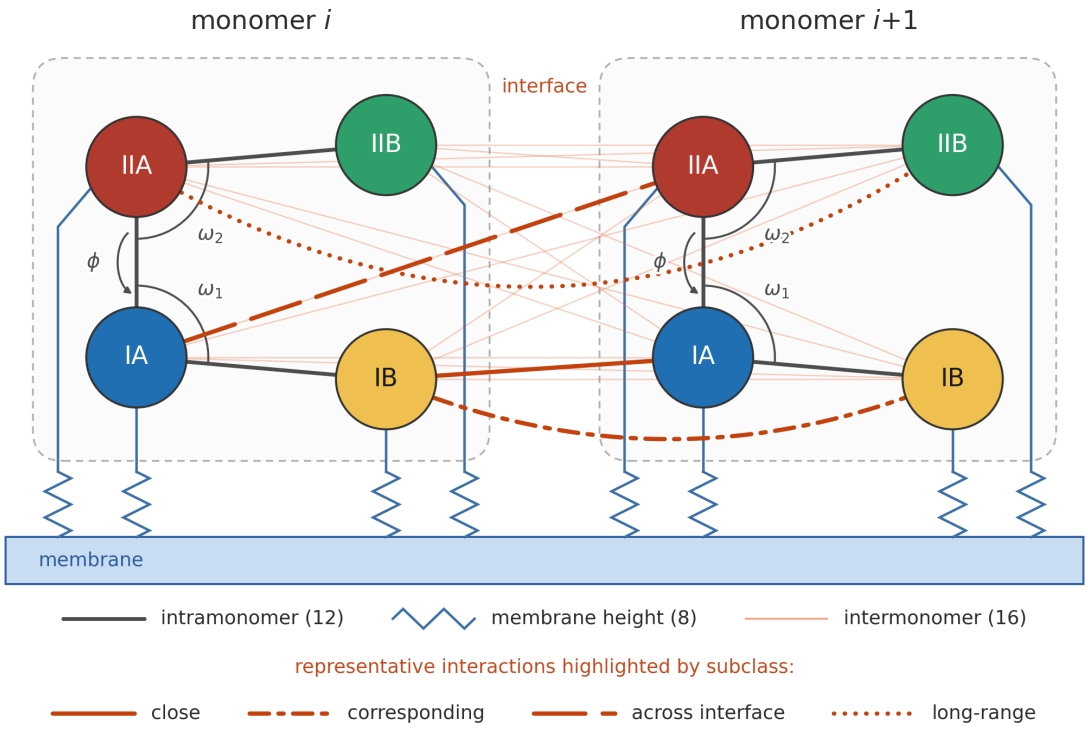


Figure 4.2: Schematic of the membrane-bound MreB^{Gs} coarse-grained dimer model. Each monomer is represented by four beads corresponding to subdomains IA, IB, IIA and IIB, with intramonomer interactions shown in grey and membrane-height restraints in blue. All 16 intermonomer interactions are shown as faint orange lines and are divided into four equal subclasses. One interaction from each subclass is additionally highlighted with a darker line to illustrate the close, corresponding, across-interface and long-range interaction types.

Figure 4.2 summarises the EICs included in the model. The 12 intramonomer EICs follow the Chu–Voth framework, with three bond lengths, two opening angles and one dihedral angle per monomer. The longitudinal interface is represented by all $4 \times 4 = 16$ possible intermonomer distances. The membrane contributes a further eight membrane-height EICs, one for each bead. The model therefore contains 36 candidate EICs in total, denoted by q_j, $j = 1, 2, \ldots, 36$. The intermonomer and membrane interaction sets are deliberately broader than the minimum required to describe the independent motion of the dimer.

To aid interpretation, the 16 intermonomer EICs are divided into four equal subclasses. *Close* interactions connect subdomains directly across the longitudinal interface, e.g. IB$^{(i)}$–IA$^{(i+1)}$. *Corresponding* interactions connect equivalent subdomains, e.g. IB$^{(i)}$–IB$^{(i+1)}$. *Across-interface* interactions connect an interface-facing subdo-

main to one further from the interface, e.g. $\mathrm{IA}^{(i)}$–$\mathrm{IIA}^{(i+1)}$. *Long-range* interactions span more widely separated subdomains across the dimer, e.g. $\mathrm{IIA}^{(i)}$–$\mathrm{IIB}^{(i+1)}$. For both the intermonomer and membrane interactions, the full candidate set is retained initially, with fluctuation matching used to determine the effective stiffnesses.

To parameterise the model, the final 40 ns of each of the five independent membrane-bound trajectories were analysed. For each EIC q_j, the mean sampled value within replica r defines the equilibrium position $q^0_{j,r}$. The magnitude of its fluctuations is then given by

$$\sigma_{j,r} = \sqrt{\left\langle \left(q_j - q^0_{j,r}\right)^2 \right\rangle_r}.$$

The replica means were averaged to give q^0_j, while $\sigma_{j,\mathrm{MD}}$ was taken as the root-mean-square of the five replica standard deviations. The value q^0_j defines the centre of the corresponding effective potential, while $\sigma_{j,\mathrm{MD}}$ is used to fit its stiffness k_j.

The harmonic treatment assumes that the analysed configurations fluctuate about a local equilibrium. As discussed in Chapter 3, the replicas do not always sample identical conformations, but their mean values remain sufficiently similar for the membrane-bound dimer to be treated within a common local harmonic description. Most EICs also show approximately stationary fluctuations over the final 40 ns, rather than sustained conformational drift. The harmonic approximation is therefore considered reasonable for this dataset, while recognising that it remains an approximation.

The present model is intended as a proof of concept and is therefore developed and validated only for the membrane-bound longitudinal dimer. In principle, the framework could be extrapolated to larger assemblies, as in the Chu–Voth actin model [45]. For $\mathrm{MreB}^{\mathrm{Gs}}$, however, this is currently limited by the absence of a resolved lateral interface. Extending only along the longitudinal interface would not reproduce the native antiparallel double-filament architecture of MreB shown in Figure 2.1a. Such a model would be of limited value for extracting biologically meaningful filament-scale properties. A more representative extension would require structural information, and ideally simulation data, for the lateral interaction between neighbouring protofilaments. The following section therefore focuses on fitting and validating the dimer model defined above.

4.2 Model Parameterisation and Validation

Having defined the coarse-grained representation and its EICs in Section 4.1, the next step is to determine the corresponding force constants, k_j, from the atomistic simulations. We first describe the fluctuation-matching procedure, which combines normal-mode analysis with iterative fitting of the force constants. We then assess the fitted model using both the target fluctuations and independent collective motions, providing a test of whether the model captures the motions of interest with sufficient accuracy.

4.2.1 Normal-mode analysis and fluctuation matching

As described in Section 4.1, the model contains 36 EICs with associated stiffnesses k_j that remain to be determined. Each EIC can be written as a function of the Cartesian coordinates, $q_j = q_j(\mathbf{x})$, where $\mathbf{x} \in \mathbb{R}^{24}$. For a small displacement $\mathbf{u} = \mathbf{x} - \mathbf{x}_0$ from the reference configuration $\mathbf{x}_0$, a first-order Taylor expansion gives

$$q_j(\mathbf{x}_0 + \mathbf{u}) \approx q_j(\mathbf{x}_0) + \nabla_{\mathbf{x}} q_j|_{\mathbf{x}_0}^T \mathbf{u}. \tag{4.2}$$

For convenience, we denote the Cartesian gradient of EIC j at the reference configuration as

$$\mathbf{b}_j = \nabla_{\mathbf{x}} q_j|_{\mathbf{x}_0} \in \mathbb{R}^{24}.$$

Using this notation, Equation (4.2) gives the change in EIC j relative to the reference configuration as

$$\Delta q_j \approx \mathbf{b}_j^T \mathbf{u}. \tag{4.3}$$

The vector $\mathbf{b}_j$ therefore describes how strongly q_j responds to small displacements of the coarse-grained sites. Stacking the gradients of all 36 EICs as rows of a matrix B gives

$$B = \begin{pmatrix} \mathbf{b}_1^T \\ \vdots \\ \mathbf{b}_{36}^T \end{pmatrix} \in \mathbb{R}^{36\times 24},$$

The gradients $\mathbf{b}_j$ were evaluated analytically for each EIC type. Let $\mathbf{r}_i = (x_i, y_i, z_i) \in \mathbb{R}^3$ denote the position of bead i. Both intramonomer bond lengths and intermonomer distances between beads i and k are given by

$$q_j = \|\mathbf{r}_i - \mathbf{r}_k\|.$$

For a membrane-referenced EIC, the coordinate is simply the bead height,

$$q_j = z_i,$$

since the membrane surface is defined at $z = 0$. The opening and dihedral angles were calculated using the geometric definitions introduced in Section 3.1.3. The Cartesian derivatives of these coordinates give the gradients $\mathbf{b}_j$.

The rank of B provides a useful check that the chosen EICs track the intended motions of the model. Since the EICs describe relative bead geometry together with bead heights above the membrane, they are unchanged by T_x, T_y and R_z. These three motions should therefore lie in the null space of B, giving an expected rank of $24 - 3 = 21$, consistent with the $3N - 3$ degrees of freedom identified in Section 4.1.

Substituting Equation (4.3) into the harmonic potential in Equation (4.1) gives

$$E_j \approx \frac{1}{2} k_j \left(\mathbf{b}_j^T \mathbf{u}\right)^2 = \frac{1}{2} \mathbf{u}^T \left(k_j \mathbf{b}_j \mathbf{b}_j^T\right) \mathbf{u}.$$

Each EIC therefore contributes a Cartesian matrix $K_j = k_j \mathbf{b}_j \mathbf{b}_j^T$. Summing the contributions from all 36 EICs gives the Cartesian Hessian, or stiffness matrix,

$$K = \sum_{j=1}^{36} K_j = B^T \operatorname{diag}(k_j) B.$$

For a given set of stiffnesses k_j, the Hessian K defines the harmonic energy associated with small displacements of the coarse-grained sites. Normal-mode analysis (NMA) decomposes this coupled motion into independent collective displacement patterns, or normal modes. Following the standard mass-weighted procedure described by Brooks et al. [46], the bead masses are collected in the diagonal matrix M, giving

$$\widetilde{K} = M^{-1/2} K M^{-1/2}.$$

Diagonalising this matrix gives

$$\widetilde{K} \mathbf{e}_\alpha = \lambda_\alpha \mathbf{e}_\alpha,$$

where $\mathbf{e}_\alpha$ describes the displacement pattern of mode α, and λ_α gives its stiffness. A displacement of the model can then be expressed as a combination of the 21 modes,

$$\mathbf{u} = \sum_{\alpha=1}^{21} a_\alpha M^{-1/2} \mathbf{e}_\alpha, \tag{4.4}$$

where a_α (the amplitude) describes how far the system has moved along mode α.

In these normal-mode coordinates, the harmonic energy is

$$E = \frac{1}{2}\sum_{\alpha=1}^{21} \lambda_\alpha a_\alpha^2.$$

For an individual mode, $E_\alpha = \frac{1}{2}\lambda_\alpha a_\alpha^2$. At thermal equilibrium, the equipartition theorem gives $\langle E_\alpha \rangle = \frac{1}{2} k_\mathrm{B} T$. Combining these expressions gives

$$\left\langle a_\alpha^2 \right\rangle = \frac{k_\mathrm{B} T}{\lambda_\alpha}, \tag{4.5}$$

where k_B is the Boltzmann constant and T is the temperature. Since the mode amplitudes describe displacements about the equilibrium structure, $\langle a_\alpha \rangle = 0$, so $\langle a_\alpha^2 \rangle$ is the variance of mode α. Thus, modes with smaller λ_α have larger thermal fluctuations than stiffer modes.

The fluctuations of all 21 modes can now be combined to determine the overall fluctuations of the coarse-grained model. These fluctuations are described by the covariance matrix $C = \langle \mathbf{u}\mathbf{u}^T \rangle$, which gives the magnitude and correlation of the Cartesian bead displacements. Since $\mathbf{u}$ is the sum of the individual mode displacements given by Equation (4.4), its covariance can be evaluated as

$$C = \left\langle \mathbf{u}\mathbf{u}^T \right\rangle = k_\mathrm{B} T \sum_{\alpha=1}^{21} \frac{M^{-1/2}\mathbf{e}_\alpha \mathbf{e}_\alpha^T M^{-1/2}}{\lambda_\alpha},$$

where the second equality follows from the mode variance in Equation (4.5). These displacements are related to changes in EIC j by $\Delta q_j \approx \mathbf{b}_j^T \mathbf{u}$. The corresponding EIC variance is therefore

$$\sigma_{j,\mathrm{CG}}^2 = \left\langle (\Delta q_j)^2 \right\rangle = \left\langle (\mathbf{b}_j^T \mathbf{u})^2 \right\rangle = \mathbf{b}_j^T \left\langle \mathbf{u}\mathbf{u}^T \right\rangle \mathbf{b}_j = \mathbf{b}_j^T C \mathbf{b}_j.$$

The resulting $\sigma_{j,\mathrm{CG}}$ can then be compared directly with the atomistic target $\sigma_{j,\mathrm{MD}}$.

Having established how to calculate $\sigma_{j,\mathrm{CG}}$, the stiffnesses k_j were fitted using the iterative fluctuation-matching procedure of Chu and Voth [45]. As a reference, each EIC was first treated as an independent harmonic coordinate, giving

$$k_j^{\mathrm{app}} = \frac{k_\mathrm{B} T}{\sigma_{j,\mathrm{MD}}^2}.$$

This is the stiffness that would reproduce the atomistic fluctuation if that EIC were uncoupled from the rest of the network. The values k_j^{app} were used as the initial estimates, $k_j^{(0)} = k_j^{\mathrm{app}}$. For each iteration, the resulting $\sigma_{j,\mathrm{CG}}^{(n)}$ was calculated by the

NMA procedure described above and compared with the atomistic target using the relative error

$$\epsilon_j^{(n)} = \frac{\sigma_{j,\mathrm{CG}}^{(n)} - \sigma_{j,\mathrm{MD}}}{\sigma_{j,\mathrm{MD}}}. \tag{4.6}$$

If the convergence criterion was not met, the stiffness was updated according to

$$k_j^{(n+1)} = \max\left[0,\, k_j^{(n)} + \eta k_j^{\mathrm{app}} \epsilon_j^{(n)}\right],$$

with $\eta = 0.05$, an arbitrary update factor. The factor k_j^{app} scales the update to the characteristic stiffness of each EIC, while the lower bound prevents unphysical negative spring constants. The procedure was repeated until the relative fluctuation error was below 0.5% for all EICs that could be further adjusted; coordinates that reached $k_j = 0$ were excluded when further agreement would require a negative stiffness. This yielded the final set of fitted stiffnesses $\{k_1, \ldots, k_{36}\}$, which are analysed in the following section.

4.2.2 Validation of the fitted model

The fluctuation-matching procedure converged after 291 iterations. The complete set of fitted coarse-grained parameters is reported in Appendix A.2. Of the 36 candidate harmonic interactions, 27 retained non-zero force constants after fitting. The fraction of these retained interactions across the intramonomer, intermonomer and membrane subclasses is summarised in Table 4.1.

Interaction subclass	**Non-zero k_j**
Intramonomer	12/12
Close interface	4/4
Corresponding	3/4
Across-interface	2/4
Long-range	1/4
Membrane-facing heights	4/4
Membrane-distant heights	1/4

Table 4.1: Number of fitted force constants that remained non-zero within each of the 36 EIC interaction subclasses after fluctuation matching. A non-zero k_j indicates that the corresponding harmonic term was retained in the final fitted model.

Table 4.1 shows that the fluctuation-matching procedure preferentially retained the more local interactions. All 12 intramonomer terms remained non-zero, preserving

the full set of internal geometric restraints. Among the intermonomer terms, all four close-interface contacts shown in Figure 4.2 retained non-zero force constants. The retained fraction then decreased as the contacts became progressively less local, from corresponding to across-interface and finally long-range interactions. This indicates that the fitted intermonomer mechanics are dominated by contacts between subdomains that are already close across the longitudinal interface.

A similar pattern was observed for the membrane restraints. All four membrane-facing heights (IA and IB) were retained, whereas only one of the four more distant heights (IIA and IIB) remained non-zero. The fitted model therefore concentrates its effective mechanical constraints on local interface contacts and the subdomains closest to the membrane. This is consistent with the strongest mechanical coupling occurring near the interface and membrane surface. However, the fitted k_j values should still be interpreted as effective parameters rather than direct measures of atomistic interaction strength.

The accuracy of the fluctuation matching was assessed using the error defined in Equation 4.6. For the 27 terms with non-zero k_j, convergence required the absolute relative error to fall below 0.5%. The remaining nine terms, with $k_j = 0$, still reproduced the atomistic fluctuations reasonably well, with errors of approximately 1–14%, comparable to Chu and Voth [45]. These relatively small errors indicate that the coordinates remain constrained through coupling to the rest of the mechanical network.

The fitted model therefore reproduces the fluctuations used during parameterisation. The next question is whether it can also reproduce independent collective motions that were not included in the fit. Of particular interest are motions describing the overall behaviour of the dimer relative to the membrane, since retaining these membrane-referenced degrees of freedom is a central feature of our new model.

A particularly relevant motion is the tilting of the dimer relative to the membrane, which is closely related to the membrane-referenced rotation R_y. To quantify this behaviour, the dimer-axis elevation θ_{tilt} was defined as the angle between the membrane and the line joining the centres of mass of the two monomers, as illustrated in Figure 4.3a. Because θ_{tilt} was not included among the 36 EIC fitting targets, it provides a held-out test of whether the fitted mechanical network reproduces larger-scale membrane-relative motion.

Unlike the EIC fluctuations used during fitting, θ_{tilt} is not itself an EIC and must instead be evaluated from the Cartesian configuration of the dimer, defined by the po-

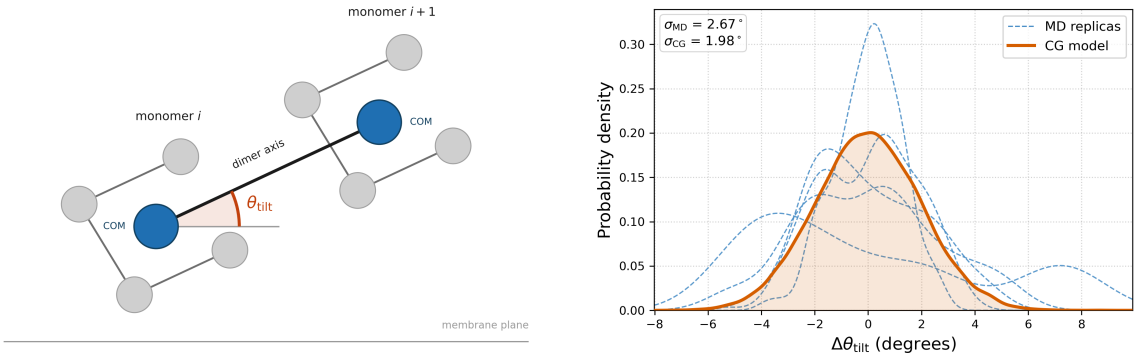
sitions of all eight beads. Configurations were therefore sampled from the equilibrium distribution of the fitted harmonic model. Using the zero-mean mode variance derived in Equation 4.5, each of the 21 non-zero normal-mode amplitudes was sampled independently according to

$$a_\alpha \sim \mathcal{N}\left(0,\ \frac{k_\mathrm{B}T}{\lambda_\alpha}\right).$$

The sampled amplitudes were then used to reconstruct a Cartesian configuration according to

$$\mathbf{x} = \mathbf{x}_0 + \sum_{\alpha=1}^{21} a_\alpha \mathbf{u}_\alpha,$$

where $\mathbf{x}_0$ is the reference configuration. The value of θ_{tilt} was then evaluated for each sampled configuration. Repeating this for 20,000 independent configurations produced the CG distribution of θ_{tilt}, from which σ_{CG} was calculated.



(a) Definition of θ_{tilt}.

(b) Fluctuations in θ_{tilt}.

Figure 4.3: Validation of the fitted coarse-grained model using the held-out dimer tilt angle θ_{tilt}. (a) Definition of θ_{tilt}. (b) Centred distributions of θ_{tilt} for the five atomistic replicas and the coarse-grained model.

Figure 4.3 shows that the coarse-grained model gives a reasonable prediction of the held-out θ_{tilt} fluctuations. To compare the local fluctuations directly, each atomistic replica was centred on its own mean before constructing the distributions. The CG distribution is smoother, as expected from the harmonic approximation, but lies well within the range sampled by the individual atomistic replicas. Quantitatively, $\sigma_{\mathrm{MD}} = 2.67°$ compared with $\sigma_{\mathrm{CG}} = 1.98°$, corresponding to an underestimate of approximately 26%. This provides useful validation that the fitted local mechanical network can generalise beyond the coordinates used during fluctuation matching and predict larger-scale membrane-relative behaviour.

Chapter 5

Discussion

This dissertation investigated how nucleotide state, polymerisation and membrane association influence the conformational and mechanical behaviour of $MreB^{Gs}$ using all-atom MD simulations. Systems representing these different conditions were first prepared and simulated. The resulting trajectories were analysed using a reduced four-subdomain description, before comparing conformational behaviour across the different simulation conditions. These atomistic results were then used to develop and validate a bespoke membrane-referenced coarse-grained model of $MreB^{Gs}$.

5.1 Key Insights

The analysis showed that the large-scale dynamics of $MreB^{Gs}$ were well captured by the relative motion of its four structural subdomains after excluding the most flexible regions. It also demonstrated the importance of independent replicas when interpreting finite MD simulations, with substantial variation observed between trajectories. This variability was large enough to obscure the comparatively subtle differences between ATP- and ADP-bound monomers. The present simulations therefore do not provide clear evidence for distinct nucleotide-dependent conformations on the timescales sampled.

Polymerisation and membrane association produced a much clearer effect, progressively restricting the conformational behaviour of $MreB^{Gs}$. Membrane binding also stabilised the membrane-facing side of the longitudinal interface, which separated in several solution replicas. Reduced mobility of the C-terminal region was identified as a likely contributor to this stabilisation. Finally, the membrane-referenced coarse-grained model reproduced the fitted atomistic fluctuations and gave a reasonable

prediction of a held-out collective variable. This suggests that it captures an effective reduced mechanical description of the membrane-bound dimer.

5.2 Future work

The atomistic simulations used five independent replicas for each condition, providing more extensive sampling than previous MreB MD studies based on only one or two trajectories. Nevertheless, the substantial variability observed between replicas indicates that further sampling would be valuable. Longer trajectories and additional replicas could test whether reproducible ATP–ADP differences emerge and whether the conclusions drawn here remain robust across a larger ensemble. In practice, however, further sampling was limited by the computational resources available.

A particularly important extension would be to simulate $MreB^{Gs}$ as a complete antiparallel double filament (Figure 2.1a). This is the characteristic filament architecture observed experimentally. At present, however, the lateral interface has not been structurally resolved for $MreB^{Gs}$, making such a model difficult to construct reliably. Once this interface is characterised, double-filament simulations could be used to investigate filament-level properties such as bending, twisting and overall mechanical behaviour, as has been done for other MreB homologues.

Resolving the lateral interface would also provide the natural next step for the coarse-grained model. The current model represents only a membrane-bound longitudinal dimer, so a complete double filament would require additional lateral interactions. These could be parameterised from atomistic double-filament simulations. The resulting model could then be extended to larger length scales and used to predict filament-level properties such as persistence length, following the multiscale approach of Chu and Voth [45].

5.3 Conclusion

This dissertation provides new insight into the conformational and mechanical behaviour of $MreB^{Gs}$ across different molecular environments. Established analysis methods were applied to obtain robust, interpretable descriptions of its dynamics. A new membrane-referenced coarse-grained model was also developed to capture its effective mechanics. Together, these results strengthen our understanding of Gram-positive MreB and provide a basis for future studies of filament-scale behaviour.

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Appendix A

Simulation and Model Parameters

A.1 Simulation Run Summary

Table A.1 summarises the MD systems simulated in this work.

System	**Ligand**	**Time (ns)**	**Atoms ($\times 10^3$)**	**Replicas**
1×1 ATP solution	ATP and Mg^{2+}	140	67	5
1×1 ADP solution	ADP and Mg^{2+}	140	67	5
2×1 ATP solution	2×ATP and 2×Mg^{2+}	140	131	5
2×1 ADP solution	2×ADP and 2×Mg^{2+}	140	131	5
2×1 ATP membrane	2×ATP and 2×Mg^{2+}	140	213	5

Table A.1: Summary of the MD simulation systems used in this work.

A.2 Coarse-Grained Model Parameters

Table A.2 reports the complete set of fitted parameters. Distance and membrane-height coordinates are reported in Å and angular coordinates in degrees, with corresponding stiffnesses k_j in $\mathrm{kcal\,mol^{-1}\,Å^{-2}}$ and $\mathrm{kcal\,mol^{-1}\,rad^{-2}}$, respectively. The dimensionless quantity χ_j compares the fitted stiffness with the apparent stiffness inferred for an independently fluctuating coordinate,

$$\chi_j = \frac{k_j}{k_j^{\mathrm{app}}}.$$

Values close to one indicate that the coordinate's own spring dominates its fluctuation, while smaller values indicate greater coupling to other coordinates.

Table A.2: Parameters of the final fitted coarse-grained model.

EIC	q_j^0	k_j	χ_j	σ_j^{MD}	σ_j^{CG}	ϵ_j (%)
Intramonomer coordinates: (+) monomer						
$d_{\mathrm{IA}-,\mathrm{IB}-}$	21.289	8.171	0.994	0.274	0.274	0.011
$d_{\mathrm{IA}-,\mathrm{IIA}-}$	24.242	16.461	0.979	0.191	0.191	0.020
$d_{\mathrm{IIA}-,\mathrm{IIB}-}$	26.731	7.356	0.986	0.287	0.288	0.143
$\omega_1^{(+)}$	87.267	545.105	0.986	1.913	1.917	0.239
$\omega_2^{(+)}$	107.722	664.439	0.978	1.725	1.726	0.031
$\phi^{(+)}$	19.615	168.440	0.998	3.462	3.463	0.025
Intramonomer coordinates: (−) monomer						
$d_{\mathrm{IA}+,\mathrm{IB}+}$	21.462	8.508	0.886	0.253	0.253	0.024
$d_{\mathrm{IA}+,\mathrm{IIA}+}$	24.121	13.323	0.986	0.213	0.214	0.011
$d_{\mathrm{IIA}+,\mathrm{IIB}+}$	26.427	8.962	0.982	0.260	0.260	0.014
$\omega_1^{(-)}$	83.913	1516.974	0.880	1.083	1.084	0.071
$\omega_2^{(-)}$	110.844	1020.615	0.857	1.303	1.304	0.038
$\phi^{(-)}$	13.212	443.251	0.972	2.106	2.106	0.003
Intermonomer interface distances						
$d_{\mathrm{IA}-,\mathrm{IA}+}$	50.655	1.573	0.556	0.467	0.467	-0.017
$d_{\mathrm{IA}-,\mathrm{IB}+}$	70.867	0.000	0.000	0.625	0.602	-3.667
$d_{\mathrm{IA}-,\mathrm{IIA}+}$	47.940	2.040	0.675	0.451	0.452	0.101
$d_{\mathrm{IA}-,\mathrm{IIB}+}$	74.146	0.000	0.000	0.560	0.510	-9.016
$d_{\mathrm{IB}-,\mathrm{IA}+}$	29.683	3.461	0.829	0.384	0.384	0.043
$d_{\mathrm{IB}-,\mathrm{IB}+}$	49.627	1.538	0.833	0.578	0.580	0.487
$d_{\mathrm{IB}-,\mathrm{IIA}+}$	30.928	2.278	0.736	0.446	0.446	0.118
$d_{\mathrm{IB}-,\mathrm{IIB}+}$	55.481	0.000	0.000	0.558	0.539	-3.438
$d_{\mathrm{IIA}-,\mathrm{IA}+}$	63.768	0.000	0.000	0.463	0.398	-13.966
$d_{\mathrm{IIA}-,\mathrm{IB}+}$	82.529	0.920	0.637	0.653	0.653	0.028
$d_{\mathrm{IIA}-,\mathrm{IIA}+}$	51.583	0.000	0.000	0.476	0.435	-8.688
$d_{\mathrm{IIA}-,\mathrm{IIB}+}$	77.243	0.000	0.000	0.668	0.638	-4.516
$d_{\mathrm{IIB}-,\mathrm{IA}+}$	45.753	1.263	0.368	0.424	0.423	-0.197
$d_{\mathrm{IIB}-,\mathrm{IB}+}$	62.164	1.025	0.655	0.628	0.627	-0.041
$d_{\mathrm{IIB}-,\mathrm{IIA}+}$	27.325	3.422	0.865	0.395	0.396	0.260
$d_{\mathrm{IIB}-,\mathrm{IIB}+}$	51.685	0.094	0.058	0.618	0.616	-0.318
Membrane-height coordinates: (+) monomer						
$z_{\mathrm{IA}}^{(+)}$	15.316	0.309	0.853	1.304	1.307	0.210
$z_{\mathrm{IB}}^{(+)}$	14.760	0.058	0.504	2.306	2.317	0.471
$z_{\mathrm{IIA}}^{(+)}$	37.537	0.000	0.000	1.069	1.017	-4.866
$z_{\mathrm{IIB}}^{(+)}$	45.261	0.000	0.000	2.127	2.106	-0.997
Membrane-height coordinates: (−) monomer						
$z_{\mathrm{IA}}^{(-)}$	15.045	0.264	0.775	1.345	1.346	0.113
$z_{\mathrm{IB}}^{(-)}$	14.983	0.195	0.235	0.862	0.859	-0.256
$z_{\mathrm{IIA}}^{(-)}$	35.933	0.000	0.000	1.905	1.721	-9.666
$z_{\mathrm{IIB}}^{(-)}$	44.075	0.535	0.899	1.017	1.017	0.047

Appendix B

Additional Results

This appendix presents additional simulation results that support the comparisons discussed in Chapter 3 but are not required for the main analysis. Figure B.1 compares the conformational distributions of ATP- and ADP-bound solution dimers. Figure B.2 compares the longitudinal interface distances of the membrane-bound ATP and solution ADP dimers.

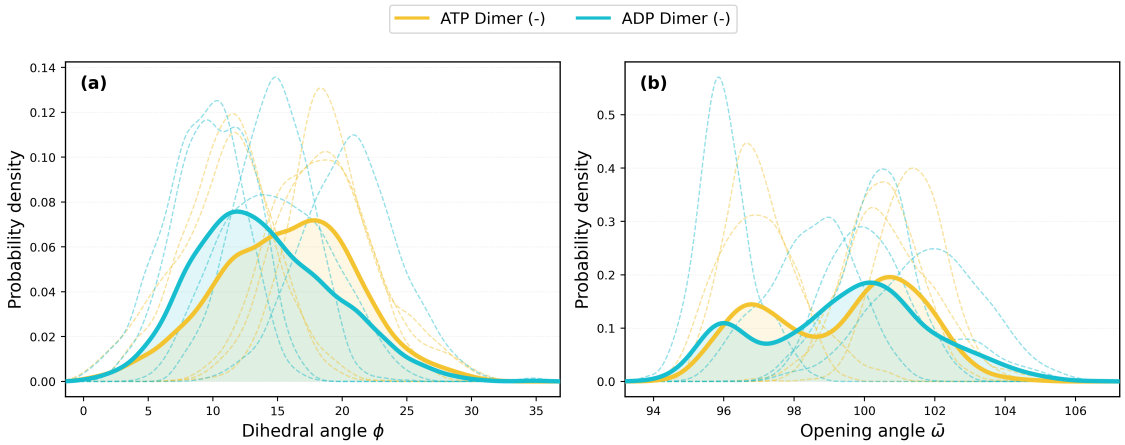


Figure B.1: Conformational distributions of ATP- and ADP-bound MreB^{Gs} solution dimers. Distributions are shown for the $(-)$ subunit of each dimer. Kernel density estimates are shown for (a) the dihedral angle ϕ and (b) the mean opening angle $\bar{\omega}$, using frames after the initial 40 ns relaxation period. Thick solid curves show the distributions pooled across all replicas for each nucleotide state, while faint dashed curves show the corresponding individual-replica distributions.

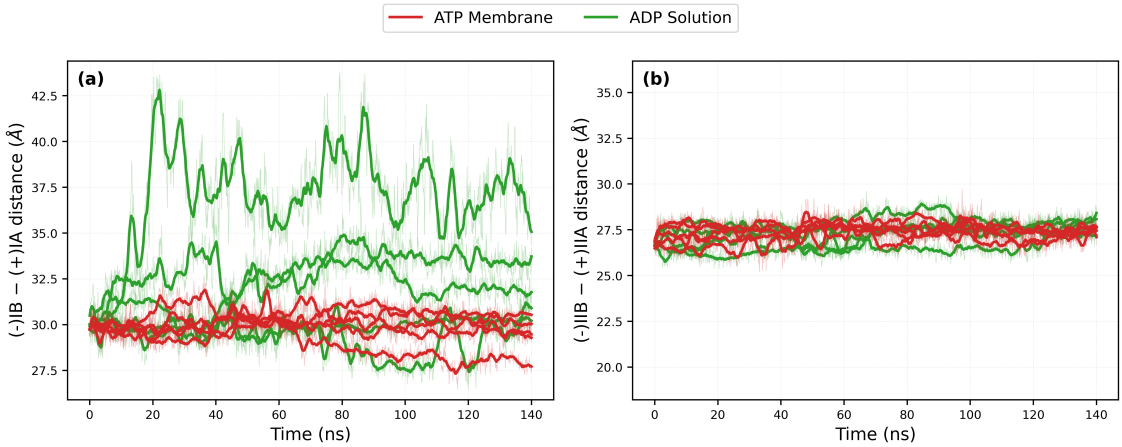


Figure B.2: Inter-subunit subdomain distances for the ATP-bound membrane-associated and ADP-bound solution $\mathrm{MreB}^{\mathrm{Gs}}$ dimers. Smoothed trajectories are shown for (a) the distance between the centres of mass of subdomains IB(−) and IA(+) and (b) the distance between subdomains IIB(−) and IIA(+). Faint lines show the corresponding unsmoothed trajectories. ATP membrane-bound replicas are shown in red and ADP solution replicas in green. Both panels span approximately 15 Å on the vertical axis to allow a fair visual comparison of the fluctuations.